Foundational Skills and Labor Market Outcomes: Evidence from India*

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Abstract

The expansion of primary schooling in developing countries has not been matched by gains in learning. This paper asks whether a program that raised foundational learning in primary school also raised earnings in early adulthood. I exploit the eligibility rule of India's District Primary Education Programme (DPEP), which targeted districts whose 1991 female literacy rate fell below the national average, in a regression discontinuity design. I observe the same birth cohorts in rural India at two stages of life: as children aged 8-11 in the Annual Status of Education Report from 2006 to 2011, and as young adults aged 22-25 in the Periodic Labour Force Survey from 2020 to 2025. DPEP eligibility raised test scores by 0.14 standard deviations, with similar gains for boys and girls. As young adults, the same cohorts earn about 32 percent more per week in regular wage or salaried jobs, but I find no earnings gains among the self-employed or casual laborers. The earnings gain is positive and significant for men but not for women. The results are robust to a range of checks. More broadly, the findings suggest that the returns to education in developing countries depend not only on how long children stay in school but also, importantly, on what they learn there.

Keywords: foundational skills; labor market outcomes; regression discontinuity design

JEL classification: I21; I25; I26; J24

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1 Introduction

Over the past three decades, low- and middle-income countries have expanded access to schooling at a remarkable pace, yet the conversion of that access into learning has lagged far behind. Enrollment is now near-universal across much of the developing world, but the competencies children acquire in school remain strikingly low. The gap between schooling and learning is visible even in the cross-section of countries. Figure 1 plots mean mathematics scores from the 2018 Programme for International Student Assessment (PISA) in relation to gross domestic product (GDP) per capita and shows that measured learning rises steeply with national income.[1] The proficiency PISA captures among fifteen-year-olds builds on basic reading and arithmetic, or foundational skills, acquired in the early grades. The international learning gap therefore reflects, in significant part, weak foundations laid in primary school. These early competencies are widely regarded as the building blocks of subsequent learning, labor market success, and human capital development more broadly.

India reflects this global pattern and is the setting for my study. Despite near-universal enrollment and substantial gains in years of schooling, foundational skills remain critically low. For example, India participated in PISA only once, in 2009, ranking seventy-second among seventy-three participating countries, and has not participated since. Moreover, the Annual Status of Education Report (ASER) has consistently documented the same shortfall for two decades: for instance, in the 2024 round, fewer than half of grade-eight students in rural areas could solve a basic division problem.[2] These deficits point to a schooling system that has struggled to translate access into actual learning.

The labor market shows a parallel imbalance. In developing countries, the share of the workforce in wage employment is low, with many workers instead engaged in self-employment or unpaid family work (Poschke, 2025), and casual work accounts for a large share of the wage employment that does exist. India reflects this pattern on a large scale: regular wage or salaried employment, the more stable and better-remunerated segment, constitutes a small share of the workforce, while casual employment remains prevalent.[3] In the PLFS 2024, about 42.7 percent of workers aged 21-60 were in wage employment: 24.2 percent in regular salaried jobs and 18.5 percent in casual wage work. In rural areas, this imbalance is even more pronounced: wage employment accounted for about 36.5 percent of workers, with 14.6 percent in regular salaried jobs and 21.9 percent in casual wage work. In addition, across Indian states, the share of the workforce in regular wage or salaried employment rises with state GDP per capita (Figure 2). This imbalance reflects the scarcity of stable, formal-sector jobs, but also a persistent mismatch between the skills employers demand and those the workforce supplies. That mismatch may originate, in part, in the weak foundational competencies documented above, on which higher-order and job-specific skills are built.

[1] PISA assesses the competencies of fifteen-year-old students in reading, mathematics, and science. Few developing countries participate, yet the results consistently reveal a positive relationship between GDP per capita and learning levels, underscoring how low learning outcomes in developing countries remain.

[2] ASER's annual reports show that children aged 5-16 have persistently low learning levels that have improved little over time.

[3] According to the Periodic Labour Force Survey (PLFS), regular wage/salaried employees are individuals working in farm or non-farm enterprises who receive a salary or wages on a regular basis, whether in the formal or informal sector.

This paper focuses on the supply side of this mismatch: the skills that workers bring to the labor market. It asks whether improving foundational learning in primary school translates into higher earnings in early adulthood in India. The question is of first-order importance. Cognitive skills measured by test scores are strong predictors of individual earnings and aggregate growth, and they often account for wage premia and cross-country income differences over and above years of schooling (Hanushek and Woessmann, 2008, 2012b; Hanushek et al., 2015, 2017). Early mastery of these skills raises productivity, facilitates subsequent learning, and improves job matching and mobility (Cunha and Heckman, 2010; Heckman, 2011). A large literature in labor economics has established the relationship between years of schooling and earnings. Years of schooling are, however, an imperfect proxy for productive human capital (Hanushek and Woessmann, 2008), and causal evidence linking early-grade learning to adult outcomes remains scarce in low- and middle-income settings.

In India, the constraint is more fundamental: no nationally representative datasets follow children into adulthood, so even the descriptive link between childhood skills and later outcomes is difficult to establish. The question is also hard to settle experimentally. One cannot randomly assign foundational skills in childhood, and the long horizon between early learning and labor market entry makes long-run randomized evidence rare. As descriptive motivation, I turn to the Young Lives Study (YLS), an Indian longitudinal survey that tracks the same children from age 8 to age 29, albeit in only two states.[4] For the older cohort, a one-standard-deviation higher reading score at age 8 is associated with a 0.36 standard deviation higher reading comprehension score at age 15, a 0.09 standard deviation higher socio-emotional index at age 22, and 0.37 log points (about 44 percent) higher weekly labor income at age 29, conditional on family background and site fixed effects. These associations, together with evidence from other countries that early cognitive gains raise adult earnings (Chetty et al., 2011; Crawfurd, 2026), make it plausible that foundational skills shape long-run labor market trajectories. On their own, however, they cannot support a causal claim.

Establishing causal evidence requires overcoming two challenges: the absence of a single dataset that measures both childhood learning and adult labor market outcomes, and the endogeneity of skill acquisition. I address both by combining a policy rule with two large surveys that observe the same birth cohorts at different stages of life. The policy is the District Primary Education Programme (DPEP), one of India's largest primary-education initiatives. Launched in 1994, DPEP targeted districts whose female literacy rate in the 1991 Census fell below the national average of 39.29 percent. Because eligibility was determined by this threshold, otherwise comparable districts on either side of it were exposed to very different levels of investment in primary schooling. I exploit this discontinuity in a regression discontinuity (RD) design.

I implement this design to two surveys. ASER measures reading and arithmetic directly, through one-on-one assessments at home, for children aged 8-11 between 2006 and 2011, who were born between roughly 1995 and 2003. The PLFS records employment and earnings for young adults aged 22-25 between 2020 and 2025, who were born over the same period. I do not observe the same individuals twice, but I observe the same cohorts in the same districts,

[4]YLS is a longitudinal study of childhood poverty coordinated by the University of Oxford. Its Indian sample follows roughly 2,000 children born in 2001-02 and 1,000 born in 1994-95 across several rounds in Andhra Pradesh and Telangana, with rich data on education, health, cognitive development, and household characteristics.

first as children in primary school and then as young adults entering the labor market. Estimating the discontinuity at the eligibility threshold in each survey reveals whether DPEP improved foundational learning and whether the cohorts it reached went on to earn more.

The design's central identifying assumption is that districts just above and just below the threshold would have had similar outcomes in the absence of the program. The institutional setting supports this assumption. The 1991 Census literacy rates were recorded before DPEP was conceived and could not have been manipulated in anticipation of it. Consistent with this, a density test in the spirit of McCrary (2008) finds no discontinuity in the distribution of districts at the cutoff.

My first result is that DPEP improved foundational learning. Children in districts just eligible for the program score 0.137 standard deviations higher on a composite measure of the ability to read a short story and to solve a division problem, the highest levels on ASER's reading and arithmetic scales. This corresponds to an increase of about 19 percent relative to the sample mean. The effect is almost identical for boys and girls. It is robust to the bandwidth selector, the use of sampling weights, the order of the local polynomial, and the use of actual program coverage rather than eligibility, with estimates ranging from 0.14 to 0.25 standard deviations.

My second result is that the same cohorts earn more in the best-paid segment of the labor market. Among young adults in regular wage or salaried employment, weekly earnings are about 32 percent higher in districts just eligible for DPEP. The gain is precisely estimated for men, at about 28 percent. For women it is at least as large in magnitude but imprecise, reflecting the small number of young women in regular jobs. The earnings gain for regular employees persists across specifications, at between roughly 15 and 41 percent. By contrast, I find no reliable effect on earnings among the self-employed or in the pooled sample, and the negative estimates for casual laborers do not survive the removal of sampling weights.

My third result concerns how much of this earnings gain runs through learning. Because the eligibility threshold is common to both surveys, the earnings effect can be expressed per standard deviation of the learning effect. The resulting ratio, about 2 log points per standard deviation, is far larger than any plausible return to foundational skills, which implies that DPEP raised earnings through other channels as well. I therefore construct a benchmark using the surrogate-index framework of Athey et al. (2019): the earnings effect DPEP would have produced if it had operated only through measured learning, valued at the skill-earnings gradient observed in the Young Lives Study. This benchmark accounts for about 18 percent of the observed effect for all persons and 22 percent for men, and for between roughly one-tenth and just over one-third (up to 44 percent for men) under plausible alternative gradients. Foundational learning is therefore an important channel, but not the only one.

These findings are effects of a program that demonstrably raised learning; they are not returns to a unit of foundational skill. DPEP was a bundled intervention that also raised enrollment and years of schooling (Azam and Saing, 2017; Khanna, 2023a), and my design captures all of these channels together. The concentration of the earnings gains in regular employment is nonetheless consistent with foundational literacy and numeracy acting as a gateway to, and a source of productivity within, stable wage work, the segment where these skills are most plausibly used and rewarded.

Several limitations qualify these estimates, and I take them up more fully in **??**. Both surveys record district of residence rather than district of schooling, so migration across the

threshold would misclassify eligibility for some individuals; because internal migration in India is low and predominantly short-distance (Munshi and Rosenzweig, 2016), this is likely to be limited and would bias my estimates towards zero. Earnings are observed only for those in paid work, and my results by employment type condition on the kind of job held, which DPEP may itself have affected. I observe these cohorts only at the start of their working lives, and my cross-district comparison does not capture general-equilibrium effects on skill prices (Khanna, 2023a).

The paper contributes to three strands of literature. First, it adds to the study of education and the labor market by providing causal evidence that improvements in foundational learning, and not only in schooling duration, are followed by higher earnings in a developing country. The micro returns to schooling attainment are well established (Becker, 1964; Mincer, 1974; Angrist and Krueger, 1991; Psacharopoulos, 1994; Card, 1999), including for India (Duraisamy, 2002; Kingdon and Theopold, 2008; Agrawal, 2011; Fulford, 2014), as are the long-run labor market effects of large school-construction programs (Duflo, 2001; Akresh et al., 2023). The macro evidence linking the quantity of schooling to growth, however, remains ambiguous (Barro, 1991; Mankiw et al., 1992; Benhabib and Spiegel, 1994; Pritchett, 2001). A leading reconciliation of this micro-macro tension is that years of schooling are an inaccurate proxy for the human capital the labor market rewards: what matters is the cognitive skills schooling is meant to impart, not time enrolled (Hanushek and Woessmann, 2008, 2012a; Pritchett, 2013; Hanushek et al., 2015, 2017; Cherry, 2020). The evidence for this view is overwhelmingly correlational, because skills are endogenous and typically observed only in cross-section (Hampf et al., 2017). It is also concentrated in high-income settings, with far less known for the low- and middle-income countries where the schooling-learning gap is widest (Glewwe, 2002; Crawfurd, 2026). I provide quasi-experimental evidence from India that a program which raised measured foundational learning was followed by substantially higher earnings in regular wage employment for the cohorts it reached.

Second, the paper contributes to the evaluation of large-scale schooling reform and to the evidence on whether such reforms raise learning rather than only access. Prior work on DPEP has focused on enrollment, retention, and completed years of schooling, with mixed conclusions about its effects on learning quality (Kumar et al., 2001; Jalan and Glinskaya, 2013; Azam and Saing, 2017). Khanna (2023a) documents its effects on schooling and wages, including through general-equilibrium adjustments in skill prices, and Roodman (2023) and Khanna (2023b) debate how robust those effects are to the construction of the district sample and to clustering. I add to this literature in two ways. Using directly measured ASER assessments for nearly 1.3 million children, I show that DPEP eligibility raised foundational competencies at the eligibility threshold, establishing that a decentralized, input-based reform improved learning and not merely attendance; this result holds with district-clustered inference and when the covered districts above the cutoff are included. Using recent PLFS rounds, I then follow the same birth cohorts into early adulthood and show where in the labor market the earnings gains appear: among regular wage and salaried employees, rather than among the self-employed or casual laborers. These findings align with evidence that interventions strengthening instructional quality, accountability, and the alignment of teaching to children's learning levels produce sizable gains in foundational skills (Banerjee et al., 2007; Muralidharan and Sundararaman, 2011; Duflo et al., 2012; Muralidharan et al., 2019). They also support the broader argument that education policy in developing countries should tar-

get learning outcomes rather than inputs alone (Hanushek, 2003; Kremer et al., 2013; World Bank, 2018).

Third, the paper offers an empirical approach for settings where the ideal longitudinal data do not exist. No Indian dataset records both childhood learning and adult labor market outcomes for the same individuals at national scale. Rather than linking individuals, I link birth cohorts and places, in the spirit of two-sample instrumental variables (Angrist and Krueger, 1992; Inoue and Solon, 2010). I apply the same eligibility rule to a learning survey that observes cohorts as children and to a labor force survey that observes them as adults. Because the policy discontinuity is defined at the district level and is fixed by a pre-program census, it can be estimated consistently in both samples. This also allows a test that is rarely possible: I observe both the program's effect on an intermediate outcome (learning) and its effect on the long-run outcome (earnings), and can therefore check how well a surrogate-index projection from learning gains alone, the approach commonly used to value education interventions (Angrist et al., 2025), recovers the observed earnings effect. The strategy can be adapted to other countries and programs where learning and adult outcomes are recorded in separate surveys and where access to schooling inputs was determined by a known assignment rule.

The remainder of the paper proceeds as follows. Section 2 describes DPEP: its objectives, funding structure, and the eligibility rule I exploit for identification. Section 3 describes the ASER, PLFS, and YLS data, the construction of the running variable, and the alignment of birth cohorts across surveys. Section 4 presents descriptive evidence from the Young Lives Study on the persistence of early skill gradients. Section 5 sets out the RD design and tests its validity. Section 6 reports the effects of DPEP eligibility on foundational learning and on early-career earnings, by employment type and gender, and Section 7 examines their sensitivity to sampling weights, polynomial order, and the use of actual program coverage. Section 8 asks how much of the earnings effect runs through foundational learning. **??** concludes.

2 The DPEP Program: Background, Objectives, and Implementation

The DPEP was launched in 1994 as a centrally sponsored scheme and was one of India's most ambitious policy initiatives to strengthen primary education and advance the goal of universal elementary education (Pandey, 2000). The program was initially rolled out across 42 districts in seven states before expanding to cover 271 educationally backward districts across 18 states by 2000 (Pandey, 2000; Azam and Saing, 2017). Over this period, DPEP served approximately 51.3 million children and led to the establishment of around 160,000 new schools, of which more than 84,000 were alternative or community-based institutions. The primary objective of DPEP was to improve access to and retention in primary and upper primary education within a holistic framework addressing the full chain from enrollment to learning quality. Its core quantitative targets were to reduce disparities in enrollment, dropout rates, and learning achievement across gender and social groups; to bring the overall primary dropout rate below 10 percent; and to raise average achievement levels by at

least 25 percent over measured baseline levels, with a minimum guarantee of foundational literacy and numeracy competencies (Pandey, 2000; Jalan and Glinskaya, 2013). In line with its decentralized design, the program emphasized district-level planning, community participation, and institutional capacity building (Pandey, 2000).

DPEP was financed through a joint funding model involving the Government of India and a consortium of international partners, including the World Bank, the European Commission, the U.K. Department for International Development, the Royal Government of the Netherlands, and the United Nations Children's Fund (UNICEF) (Jalan and Glinskaya, 2013). By 2002, the World Bank alone had committed approximately $1.62 billion to the program, while other donors concentrated on specific states; in 2002-03 alone, $345 million in foreign funds was spent in fewer than half of the country's districts (Khanna, 2023a). Crucially, DPEP targeted districts with female literacy rates below the national average of 39.29 percent recorded in the 1991 Census, thereby explicitly prioritizing regions with the deepest educational deficits and the sharpest gender disparities (Azam and Saing, 2017; Khanna, 2023a).

DPEP was subsumed into Sarva Shiksha Abhiyan (SSA), India's national program for universal elementary education, in the early 2000s. This matters for interpretation. The earliest cohorts in my samples attended primary school while DPEP was in operation, whereas the latest cohorts entered school after it had been absorbed into SSA. For the later cohorts, the eligibility effect therefore captures the persistent schooling infrastructure and institutions created by DPEP, together with any differences in subsequent SSA support.

I exploit this targeting rule to identify the causal effect of DPEP on foundational skills and early-career earnings. Following Khanna (2023a), I use the district-level eligibility threshold based on the 1991 female literacy rate. Districts with female literacy rates below 39.29 percent were eligible for DPEP, and treatment status closely follows this rule.[5] Because the program was largely allocated according to the eligibility criterion, my main analysis uses a sharp RD design, comparing individuals in districts just below and just above the cutoff.

3 Data

The analysis combines two nationally representative household surveys that observe the same birth cohorts at different stages of life. The Annual Status of Education Report (ASER) measures these cohorts' foundational learning during primary school, while the Periodic Labour Force Survey (PLFS) records their employment and earnings as young adults. I link the two surveys using each individual's district-level DPEP eligibility status, determined by the district's female literacy rate in the 1991 Census. The sampling designs and survey procedures are described in Subsection A.1 and Subsection A.2. Here, I explain how I construct the variables used in the analysis.

ASER assesses each child aged 5-16 in sampled rural households individually at home, regardless of whether the child is enrolled in school. The reading and arithmetic tests use ordinal scales. The assessor moves the child through a fixed sequence of tasks, and the

[5]Although eligibility was restricted to districts with female literacy rates below the national average, four districts above the cutoff also received the program (Khanna, 2023a; Roodman, 2023). My main analysis excludes these four districts; Subsection 7.3 re-includes them in a fuzzy design.

survey records only the highest level the child completes. Reading has five levels: cannot read, can recognize letters, can read words, can read a short paragraph at Grade 1 difficulty, and can read a short story at Grade 2 difficulty. Arithmetic also has five levels: cannot recognize numbers, can recognize numbers from 1 to 9, can recognize numbers from 11 to 99, can perform two-digit subtraction with borrowing, and can perform three-digit-by-one-digit division.[6] Because the levels are nested, a child recorded at a given level can perform all tasks at lower levels. The assessment tools remained unchanged across the rounds used in my analysis, ensuring that the recorded levels are comparable over time.

The ordinal nature of these scales guides my approach to measuring skills. The level codes have no cardinal meaning: the step from letters to words is not equivalent to the step from a paragraph to a story, and averaging the codes would impose arbitrary spacing between levels. I therefore summarize each scale using a binary indicator for reaching its highest level. Read-Story equals one if the child can read the Grade 2 story, and Math-Division equals one if the child can solve the division problem. These are competencies children are expected to have acquired by ages 8-11. I combine the two indicators into a composite Test Score, defined as their sum, which ranges from 0 to 2. For estimation, I standardize the composite using the pooled sample's mean and standard deviation so that effects are expressed in standard-deviation units.

I pool the six ASER rounds from 2006 to 2011 and restrict the sample to children aged 8-11. The sample comprises 1,298,036 children, including 703,826 boys and 594,210 girls. Foundational skills among these children are low. Table A2 reports the full distribution across proficiency levels by survey round. Only 35-39 percent of children aged 8-11 could read a Grade 2-level story, while the share who could solve a division problem declined from 30 percent in 2006 to 19 percent in 2011. The mean composite score is 0.61 (standard deviation 0.82), with mean scores of 0.62 for boys and 0.59 for girls, as shown in Table 2.

I pool the PLFS rounds from 2020 to 2025 and restrict the sample to individuals aged 22-25 in rural India. I define activity status using the current weekly status, which refers to the seven days preceding the survey, because the PLFS earnings questions are tied to this reference period. Workers are classified into three mutually exclusive categories: self-employed, regular wage or salaried employees, and casual wage laborers. The PLFS records earnings differently across employment types and reference periods, so I convert them into a common measure of weekly earnings. For casual laborers, weekly earnings are the sum of daily wage earnings from principal and subsidiary activities over the seven days of the reference week. For regular wage and salaried employees, weekly earnings are calculated by multiplying earnings in the preceding calendar month by 7/30. For the self-employed, weekly earnings are calculated by multiplying self-employment earnings over the preceding 30 days by 7/30. I deflate weekly earnings to constant prices using a year-specific price index (CPI-Rural, base 2012 = 100), winsorize real earnings at the 1st and 99th percentiles, and use the natural logarithm of one plus real weekly earnings as the outcome.

The PLFS sample contains 79,548 individuals in the labor force, including 59,551 men and 19,993 women; 66,749 of them are employed, while the remaining individuals are unemployed

[6]The 2006 round did not separately assess recognition of numbers 1-9, so the two lowest arithmetic categories are combined in that year. This does not affect my outcome measures, which are defined at the top of each scale.

and assigned zero earnings. Of these, 76,788 can be matched to districts with 1991 literacy data and are included in the estimation.[7] Table 3 reports summary statistics. Mean real weekly earnings, including zeros for the unemployed, are INR 777, with substantial variation across employment types: INR 1,577 for regular wage or salaried employees, INR 1,153 for casual laborers, and INR 548 for the self-employed. Average weekly earnings are INR 908 for men and INR 385 for women. The gender gap persists across employment categories: INR 1,620 versus INR 1,377 among regular employees, INR 1,201 versus INR 766 among casual laborers, and INR 711 versus INR 220 among the self-employed. Because earnings are observed only for those in the labor force, and DPEP may have influenced both labor force participation and the type of employment individuals enter, the earnings regressions condition on outcomes that the program itself may have affected.

DPEP eligibility was determined by a district's female literacy rate in the 1991 Census. Districts with female literacy rates below the 1991 national average of 39.29 percent were eligible for the program (Azam and Saing, 2017; Khanna, 2023a). I define the running variable as $\tilde{R}_d = c - R_d$, where R_d is the 1991 female literacy rate of district d and $c = 39.29$. This definition places eligible districts to the right of zero in all figures and tables. I harmonize district identifiers across both surveys to 2001 Census boundaries and match each district to its 1991 female literacy rate. Because the running variable is defined at the district level, multiple individuals share the same value. I therefore cluster standard errors at the district level, which is also the level at which eligibility was assigned, and account for mass points in the running variable during estimation.

The two samples cover the same birth cohorts. Children aged 8-11 in the ASER rounds from 2006 to 2011 were born between approximately 1995 and 2003, while PLFS respondents aged 22-25 in the rounds from 2020 to 2025 were born over the same period. Figure 3 maps each survey round to the birth cohorts it covers. All these cohorts entered primary school after DPEP was launched in 1994. Because birth year is inferred from reported age and survey year, the alignment is approximate. ASER and the PLFS are independent surveys: I do not observe the same individuals in both, but I observe individuals from the same birth cohorts in the same districts. Both surveys record the district of current residence rather than the district where individuals attended school. Internal migration in India is limited and predominantly short-distance (Munshi and Rosenzweig, 2016); migration across the eligibility threshold would misclassify some individuals' eligibility status.

The ASER and PLFS samples link childhood learning to adult earnings at the level of birth cohorts and districts, not individuals. To examine the individual-level relationship between early skills and later outcomes, I also use the Young Lives Study (YLS), a longitudinal survey that follows the same children from primary school into adulthood. Coordinated by the University of Oxford, the YLS follows two cohorts of children in Andhra Pradesh and Telangana. I use the older cohort, comprising approximately 1,000 children born in 1994-95. These children were first surveyed at age 8 in 2002 and re-interviewed at ages 15 (2009), 22 (2016), and 29 (2023-24). Subsection A.3 describes the survey in detail.

From the YLS, I construct a standardized reading score at age 8 as my measure of foundational literacy. This measure is comparable to the ASER reading assessment, although no comparable ASER-style mathematics assessment is available for eight-year-olds in the

[7] I exclude Jammu and Kashmir because the 1991 Census was not conducted in the state.

YLS. I relate foundational literacy at age 8 to outcomes measured at three later stages of life. At age 15, I measure reading comprehension using a sentence-completion (CLOZE) test and receptive vocabulary using the Peabody Picture Vocabulary Test (PPVT). At age 22, I use self-esteem and socio-emotional indices, constructed via polychoric principal component analysis from 31 and 26 self-reported items, respectively. At age 29, I use the logarithm of one plus weekly labor income, combining cash and in-kind payments converted to weekly equivalents. All scores are standardized within the sample. Because the YLS sites were purposively selected rather than randomly sampled, the data are not representative of India, and the associations they reveal cannot be interpreted as causal. I use the YLS for two limited purposes: to document the persistence of the relationship between early skills and later outcomes over the life course in Section 4, and to estimate the skill-earnings relationship used in the surrogate-index benchmark in Subsection 8.2.

4 Motivating Evidence: Foundational Skills and Later-Life Outcomes

I begin with descriptive evidence on whether foundational skills acquired in childhood are associated with later-life outcomes, using the YLS older cohort observed at ages 8 (2002), 15 (2009), 22 (2016), and 29 (2023-24). I estimate

$$Y_{ic} = \alpha + \gamma\, S_{ic} + X_{ic}'\delta + \mu_c + \varepsilon_{ic}, \tag{1}$$

where Y_{ic} is an outcome for child i in sentinel site c, and S_{ic} is the standardized reading score at age 8. X_{ic} is a vector of baseline controls: the child's sex, age in months, household size, caste, religion, whether the primary caregiver completed primary schooling, and the household wealth index. μ_c denotes sentinel-site fixed effects. Standard errors are clustered at the sentinel-site level. The outcomes are standardized cognitive test scores at age 15, standardized non-cognitive indices at age 22, and the log of one plus weekly labor income at age 29. Table 1 reports the estimates.

The estimates reveal a strong and persistent gradient across the life course. A one-standard-deviation increase in the age-8 reading score is associated with a 0.356 standard deviation higher score on the CLOZE reading comprehension test and a 0.296 standard deviation higher score on the PPVT at age 15, both significant at the 1 percent level. The association extends to non-cognitive outcomes at age 22, although it is weaker and less uniform. A one-standard-deviation increase in the age-8 reading score is associated with a 0.086 standard deviation increase in the socio-emotional index, which captures conscientiousness, emotional stability, and agency, and is significant at the 1 percent level. By contrast, the association with self-esteem is essentially zero after controlling for background characteristics: the point estimate is -0.005 and statistically insignificant.

Most importantly, the gradient persists into the labor market. A one-standard-deviation increase in the age-8 reading score is associated with 0.368 log points higher weekly income at age 29, significant at the 1 percent level, so that a 0.1 standard deviation increase in early reading corresponds to roughly 3.7 percent higher earnings. The gradient remains large and precisely estimated conditional on sex, caste, religion, caregiver education, wealth, and site

fixed effects.[8] Foundational skills are thus associated with adult economic outcomes beyond what is captured by family circumstances at age 8.

These results are suggestive but not conclusive. First, they are correlational. The controls and site fixed effects absorb observable differences in family background and local conditions, but children who read well at age 8 may differ from those who do not in unobserved ability, parental motivation, or early childhood investments, any of which could independently improve later outcomes. Second, the YLS sample is drawn from purposively selected sites in a single region (undivided Andhra Pradesh) and is not representative of India. The remainder of the paper addresses these limitations by turning to large surveys of learning (ASER) and labor market outcomes (PLFS) and exploiting the discontinuity in DPEP eligibility at the 1991 female literacy threshold.

5 Empirical Strategy

DPEP eligibility was determined by a deterministic rule: districts with a 1991 Census female literacy rate below the national average of 39.29 percent were eligible. Let R_d denote the 1991 female literacy rate of district d and $c = 39.29$ the cutoff. As described in Section 3, I recenter and sign the running variable as $\tilde{R}_d = c - R_d$, so that eligible districts lie to the right of zero, and define eligibility as $D_d = \mathbb{1}\{\tilde{R}_d \geq 0\}$. I estimate the effect of eligibility using a sharp regression discontinuity design:

$$Y_{id} = \alpha + \tau D_d + \beta_1 \tilde{R}_d + \beta_2 \, D_d \times \tilde{R}_d + \varepsilon_{id}, \qquad |\tilde{R}_d| \leq h, \tag{2}$$

where Y_{id} denotes the outcome for individual i in district d. In the ASER sample, Y_{id} is the standardized composite test score of a child aged 8-11. In the PLFS sample, it is the log of one plus weekly real earnings of a young adult aged 22-25. The interaction term allows the relationship between the outcome and the running variable to differ on either side of the cutoff. The coefficient τ measures the discontinuity in the conditional mean of the outcome at the threshold.

I estimate Equation 2 by local linear regression with a triangular kernel, implemented in the `rdrobust` package (Calonico et al., 2014). Each observation receives a weight equal to the product of its survey sampling weight and its kernel weight. Because the running variable is defined at the district level, many individuals share the same value. I therefore apply the mass-point adjustment in `rdrobust` and cluster standard errors at the district level, where eligibility is assigned. I report estimates using two data-driven bandwidth selectors. The MSE-optimal bandwidth minimizes the asymptotic mean squared error of the point estimator, while the CER-optimal bandwidth minimizes the coverage error of the robust bias-corrected confidence intervals (Calonico et al., 2020). By construction, the CER-optimal bandwidth is narrower.[9] The tables report conventional point estimates with

[8] Table A1 re-estimates the earnings specification separately for men and women. For men, a one-standard-deviation increase in the age-8 reading score is associated with 0.387 log points higher weekly income at age 29, significant at the 1 percent level and close to the pooled estimate of 0.368. For women, the point estimate is smaller, at 0.217, but it is imprecisely estimated and statistically insignificant. The female sample is also smaller, with 252 observations compared with 381 for men.

[9] Subsection A.4 describes the MSE- and CER-optimal bandwidth selectors.

district-clustered standard errors.

Identification rests on the assumption that the conditional expectations of potential outcomes are continuous in $\tilde{R}_d$ at the cutoff. Under this assumption, τ identifies the intention-to-treat (ITT) effect of DPEP eligibility for districts at the threshold. Two features of this estimand matter for interpretation. First, τ measures the effect of eligibility, not program receipt. A few districts above the cutoff were also covered; Subsection 7.3 shows that my estimates are robust to including them through a fuzzy design. Second, DPEP bundled several primary-school inputs, including new schools, additional teachers, and learning materials. The effect of eligibility on adult earnings therefore reflects every channel through which the program operated, including improvements in foundational learning, additional years of schooling, and possibly changes in local economic conditions. Section 8 returns to this point.

5.1 Validity of the Design

The central threat to an RD design is that units sort precisely around the cutoff, so that districts just above and just below the threshold differ systematically. Because the running variable is defined at the district level, I test for sorting by examining whether the density of district female literacy rates is continuous at the cutoff. I implement the manipulation test of McCrary (2008) using the local polynomial density estimator of Cattaneo et al. (2018, 2020) with a triangular kernel. Figure 4 plots the estimated density of $\tilde{R}_d$ on either side of the threshold, with 95 percent confidence bands. The estimated densities are similar at the cutoff, and the confidence bands overlap substantially. The formal test yields a robust T-statistic of 0.701 (p-value = 0.484), so I cannot reject the null hypothesis that the density of district female literacy rates is continuous at the cutoff.

The absence of sorting is consistent with the institutional setting. The running variable is the district female literacy rate from the 1991 Census, which was enumerated before DPEP was designed in 1993-94 and launched in 1994. Districts could not have influenced their recorded literacy rates in anticipation of the program, since neither the program nor its eligibility rule existed when the data were collected. The cutoff itself, the national average female literacy rate, was determined by the same Census. A district's position relative to the threshold therefore reflects pre-existing differences in female literacy rather than a strategic response to the program.

6 RD Results

6.1 DPEP Eligibility and Foundational Skills

I first examine whether DPEP eligibility improved the foundational learning of children who attended primary school after the program began. Table 4 reports sharp RD estimates for the standardized composite test score in the pooled ASER sample of 1,298,036 children aged 8-11, observed between 2006 and 2011. The composite is the sum of two binary indicators, Read-Story and Math-Division, which represent the highest levels on ASER's reading and arithmetic scales. The estimates therefore reflect foundational learning broadly rather than

skills in a single subject. Column (1) reports estimates for all children, and columns (2) and (3) for boys and girls. Within each panel, the subgroup estimates use the bandwidth selected for the full sample, so all three columns are estimated over a common window.

Panel A reports estimates using the MSE-optimal bandwidth of 11.72 percentage points of 1991 female literacy. Children in districts just eligible for DPEP score 0.137 standard deviations higher than children in districts just ineligible, an estimate significant at the 5 percent level and based on 550,344 children within the bandwidth. Panel B uses the narrower CER-optimal bandwidth of 8.54 percentage points, which reduces the effective sample to 364,793 children. The estimated discontinuity rises slightly to 0.152 standard deviations and remains significant at the 5 percent level. The stability of the estimate in the narrower window indicates that the result is not driven by observations far from the threshold.

Standard errors throughout are clustered at the district level because treatment is assigned by district, not by child. Clustering also helps explain why the selected bandwidths are relatively wide. Because each district effectively contributes one unit of independent variation, the data contain far fewer independent observations than the 1.3 million children suggest, and the optimal bandwidth must be wide enough to include enough districts on each side of the cutoff. Without clustering, the procedure selects a much narrower bandwidth, but that would implicitly treat assignment as occurring at the individual level.[10]

The effects are similar for boys and girls. With the MSE-optimal bandwidth, eligibility raises test scores by 0.141 standard deviations for boys, significant at the 5 percent level, and by 0.133 standard deviations for girls, significant at the 10 percent level. The CER-optimal estimates follow the same pattern: 0.159 standard deviations for boys and 0.144 for girls. The point estimates for the two groups differ by less than 0.02 standard deviations in either panel, a difference small relative to the standard errors. Although DPEP targeted districts based on female literacy and emphasized reducing gender gaps in schooling, its effect on foundational learning at the threshold was shared by boys and girls. The ASER results establish the first link in my argument: cohorts exposed to DPEP eligibility in primary school acquired measurably stronger foundational skills.

6.2 DPEP Eligibility and Early-Career Earnings

I next examine whether the same birth cohorts, now aged 22-25, earn more in the labor market. Table 5 reports sharp RD estimates for the log of one plus weekly real earnings in the pooled PLFS sample, observed between 2020 and 2025. Columns (1)-(4) report estimates for all persons, columns (5)-(8) for men, and columns (9)-(12) for women. Within each group, the first column covers everyone in the labor force (current weekly status codes 11 to 81), including the unemployed, who are assigned zero earnings, and the remaining three columns report estimates separately for the self-employed, regular wage or salaried employees, and casual wage laborers. Within each panel, all columns use the bandwidth selected for the full sample, so they are estimated over a common window.

Panel A reports estimates using the MSE-optimal bandwidth of 9.46 percentage points of 1991 female literacy, and Panel B the narrower CER-optimal bandwidth of 6.89 percentage points. The estimation sample contains 76,788 individuals from 313 districts, of whom 24,805

[10] Appendix B reports estimates with unclustered standard errors and correspondingly reselected bandwidths.

lie within the MSE-optimal bandwidth. Standard errors are clustered at the district level.

For the labor force as a whole, the estimated discontinuity in the log of one plus earnings is −0.172 with the MSE-optimal bandwidth and −0.267 with the CER-optimal bandwidth. Neither estimate is statistically distinguishable from zero, and the confidence intervals include both large positive and large negative effects. The imprecision reflects the heterogeneity of the pooled sample: mean weekly earnings range from INR 548 among the self-employed to INR 1,577 among regular employees, earnings among the self-employed are highly dispersed, and the unemployed contribute zeros. The pooled estimate also conflates any effect of DPEP on earnings within a given type of work with any effect on the composition of employment and on the probability of being employed. Because the outcome is the log of one plus earnings and includes zeros, the pooled estimate should not be read as a percentage change. I therefore examine the employment categories separately.

The clearest result concerns regular wage and salaried employees, the most stable and best-paid segment of the labor market for young adults. Eligibility raises log weekly earnings among regular employees by 0.281 with the MSE-optimal bandwidth, an estimate significant at the 5 percent level and based on 5,289 regular employees within the bandwidth. Regular employees in districts just eligible for DPEP therefore earn about 32 percent more per week than their counterparts in districts just ineligible. The CER-optimal estimate is very similar, at 0.259, or about 30 percent, and is also significant at the 5 percent level.

The effect for regular employees is present for both men and women. For men, eligibility raises log weekly earnings by 0.248 with the MSE-optimal bandwidth, or about 28 percent, an estimate significant at the 5 percent level; the CER-optimal estimate is 0.223 and is significant at the 10 percent level. For women, the point estimate is larger, at 0.336, but it is based on only 963 women within the bandwidth and is significant only at the 10 percent level. With the CER-optimal bandwidth, the estimate for women is 0.305 and is not statistically significant. The female estimates are consistent with an effect similar to, or larger than, that for men, but the data do not allow me to distinguish the female estimate precisely from either the male estimate or zero.

For casual wage laborers, the estimated discontinuity is negative. Eligibility lowers log weekly earnings among casual laborers by 0.160 with the MSE-optimal bandwidth, or about 15 percent, significant at the 5 percent level; the CER-optimal estimate is −0.211, also significant at the 5 percent level. The negative estimates are larger for women: −0.334 (significant at the 10 percent level) and −0.406 (significant at the 5 percent level), although they are based on only 722 and 503 women within the respective bandwidths. For men, the MSE-optimal estimate is −0.105 and not significant, and the CER-optimal estimate is −0.141, significant at the 10 percent level. As Subsection 7.1 shows, the casual-labor estimates are much less robust than those for regular employees: they shrink substantially and lose significance when the sampling weights are dropped.

For the self-employed, the estimates are uninformative in every specification. With the MSE-optimal bandwidth, the discontinuity is −0.226 for all self-employed workers, −0.512 for men, and 0.139 for women. Self-employment among young adults in rural India spans unpaid work in family enterprises, own-account farming, and small nonfarm businesses. Reported earnings in this category are highly dispersed and frequently close to zero, which makes the discontinuity difficult to estimate precisely; because the outcome is the log of one plus earnings, these coefficients are also not close to percentage changes.

Taken together, Table 4 and Table 5 establish the second link in my argument: cohorts exposed to DPEP eligibility in primary school acquired stronger foundational skills as children and earn substantially more as young adults in regular wage employment. Three considerations qualify the interpretation of the earnings estimates.

First, the estimates are reduced-form effects of program eligibility, not returns to foundational skills. The ASER and PLFS samples cover the same birth cohorts in the same districts, but not the same individuals. More importantly, DPEP increased years of schooling as well as learning (Azam and Saing, 2017; Khanna, 2023a), and it may also have affected local labor markets. The earnings discontinuities therefore combine the effects of improved foundational learning with those of additional schooling and any other channel through which the program operated. Section 8 asks how much of the earnings effect measured learning can account for.

Second, the estimates within employment categories condition on the type of work an individual holds, which DPEP may itself have affected. If eligibility moved some young adults out of casual labor and into regular employment, the composition of both groups would change at the cutoff. Suppose those who moved were more skilled than the casual laborers who remained but less skilled than existing regular employees. The change in composition would then lower average earnings in both groups. Under this pattern, the positive estimate for regular employees would understate the effect for a fixed population of regular workers, while the negative estimate for casual laborers could reflect selection out of casual work rather than a fall in casual wages. The earnings estimates alone cannot distinguish between these explanations. The results for regular employees are best interpreted as the earnings difference among those holding regular jobs at the threshold, reflecting both higher productivity within these jobs and any change in who holds them.

Third, the sample is conditional on labor force attachment. Individuals outside the labor force are excluded, while the unemployed enter the pooled estimates with zero earnings. This restriction matters most for women, who account for only one in four individuals in my labor-force sample; only 2,647 women hold regular employment. If DPEP affected women's labor force participation, the estimates for women would be conditional on a selected group. Together with the small sample sizes, this selection is why I treat the female estimates as suggestive.

7 Robustness

This section examines whether the main results depend on the use of sampling weights, the order of the local polynomial, or the treatment of eligibility rather than actual program coverage. Each check is reported for both the ASER test-score estimates and the PLFS earnings estimates. The comparison of MSE- and CER-optimal bandwidths, already reported in Panels A and B of each main table, shows sensitivity to the estimation window.

7.1 Unweighted Estimates

My main estimates use survey weights so that they represent the rural population of children and young adults in districts near the cutoff. Weighting, however, can increase variance and

give disproportionate influence to a few observations with large multipliers. Table A3 and Table A7 re-estimate the main specifications without sampling weights, reselecting the MSE- and CER-optimal bandwidths for the unweighted sample.

For foundational skills, the unweighted estimates are larger than the weighted estimates. With the MSE-optimal bandwidth of 8.51, eligibility raises the standardized test score by 0.185 standard deviations for all children, 0.184 for boys, and 0.186 for girls, all significant at the 5 percent level. With the CER-optimal bandwidth of 6.20, the estimates are 0.164, 0.157, and 0.171 standard deviations, respectively, significant at the 10 percent level. Without weights, the estimate for girls is as precise as that for boys, reinforcing the conclusion that the learning gains were shared equally.

For earnings, the unweighted results confirm the central finding for regular employees. With the MSE-optimal bandwidth of 10.00, eligibility raises log weekly earnings among regular employees by 0.199 for all persons, significant at the 5 percent level. The corresponding estimates are 0.168 for men (significant at the 5 percent level) and 0.249 for women (significant at the 10 percent level). With the CER-optimal bandwidth, the estimate for all regular employees is 0.171, significant at the 10 percent level, while the estimates for men (0.132) and women (0.214) remain positive but are no longer statistically significant. The unweighted earnings gains for regular employees are therefore somewhat smaller than the weighted estimates, at roughly 19 to 22 percent rather than 30 to 32 percent, but they have the same sign and a similar order of magnitude.

By contrast, the negative effect on casual earnings does not persist when sampling weights are removed. The unweighted MSE-optimal estimate for all casual laborers is -0.025, and the estimate for women is essentially zero, at 0.015. None of the unweighted casual-labor estimates is statistically significant. The weighted casual-labor results thus appear to be driven by observations with large sampling weights rather than by a discontinuity present throughout the sample. The estimates for the labor force as a whole and for the self-employed remain imprecise and, with one exception, statistically insignificant: the CER-optimal estimate for women in the labor force (-0.683) is significant at the 10 percent level.

7.2 Local Quadratic Polynomial

The main estimates approximate the conditional mean of the outcome on each side of the cutoff with a linear function. Table A4 and Table A8 instead fit local quadratic polynomials, with and without sampling weights, using MSE-optimal bandwidths selected for the quadratic specification. A quadratic fit reduces bias from curvature in the conditional mean but increases variance, so larger standard errors are expected.

For foundational skills, the quadratic estimates closely match the linear estimates. With sampling weights and a bandwidth of 13.92, eligibility raises test scores by 0.160 standard deviations for all children, 0.167 for boys, and 0.151 for girls, each significant at the 10 percent level. Without weights, the estimates are 0.195, 0.192, and 0.197 standard deviations, each significant at the 5 percent level.

For earnings, the weighted quadratic estimates for regular employees are almost identical to the linear estimates: eligibility raises log weekly earnings by 0.299 for all regular employees and by 0.257 for men (both significant at the 5 percent level). For women, the point estimate is 0.356 but is not statistically significant. Without weights, the estimate for all

regular employees falls to 0.144 and is no longer statistically significant, although its sign and magnitude are close to those of the unweighted linear estimate of 0.199. With weights, the negative casual-labor estimates persist under the quadratic specification (−0.181 for all casual laborers and −0.413 for women); without weights, they again become small and statistically insignificant. The estimates for the labor force as a whole and for the self-employed remain imprecise throughout; the one marginally significant estimate (−0.972 for women in the labor force, unweighted) is not significant in any weighted specification.

7.3 Program Coverage Rather Than Eligibility

The main estimates measure the effect of DPEP eligibility. Program coverage did not follow the eligibility rule exactly: four districts above the cutoff were also covered by DPEP (Khanna, 2023a; Roodman, 2023). To recover the effect of program coverage, I estimate a fuzzy RD in which an indicator for DPEP coverage is instrumented by eligibility. The coverage indicator equals one for districts below the national average and for the four covered districts above it. The fuzzy estimate is the ratio of the discontinuity in the outcome to the discontinuity in coverage at the cutoff. Under monotonicity and the exclusion restriction, it identifies the effect of DPEP coverage for complier districts at the threshold. I use the MSE-optimal bandwidths from the corresponding sharp RD specifications.

Table A5 reports the results for test scores. In the ASER sample, the discontinuity in DPEP coverage at the cutoff is 0.743 with sampling weights and 0.730 without them, both significant at the 1 percent level, so the first stage is strong. Because coverage jumps by less than one, the fuzzy estimates scale up the sharp estimates. With sampling weights, DPEP coverage raises test scores by 0.185 standard deviations for all children, 0.189 for boys, and 0.180 for girls, each significant at the 10 percent level. Without weights, the corresponding estimates are 0.253, 0.250, and 0.256 standard deviations, each significant at the 5 percent level.

Table A9 reports the results for earnings. In the PLFS sample with sampling weights, the discontinuity in coverage ranges from 0.70 to 0.81 across employment groups, except for female casual laborers, for whom it is 0.592; it is significant at the 1 percent level in every column. For regular employees, DPEP coverage raises log weekly earnings by 0.347 for all persons (significant at the 1 percent level), 0.306 for men (5 percent level), and 0.417 for women (10 percent level), corresponding to earnings gains of about 36 to 52 percent in complier districts. Without sampling weights, the effects for regular employees remain positive and significant: 0.249 for all persons and 0.207 for men (both at the 5 percent level), and 0.325 for women (10 percent level). As in the sharp RD, the estimates for casual laborers are negative and marginally significant with weights (−0.229 for all casual laborers) but close to zero and insignificant without them.

In summary, the robustness checks yield a consistent picture. The effect of DPEP on foundational learning is positive and statistically significant under both bandwidth selectors, with and without sampling weights, under linear and quadratic specifications, and when eligibility is replaced by actual coverage. Its magnitude ranges from 0.137 to 0.253 standard deviations, and the effects for boys and girls are similar in every specification. The earnings gain for regular wage and salaried employees is positive in every specification and statistically significant for all persons in all but one, the unweighted local quadratic specification, in which

the point estimate remains close to its linear counterpart. Its magnitude ranges from about 15 to 41 percent, with the estimates for men more precise than those for women. By contrast, the negative estimates for casual laborers depend on the use of sampling weights, and the estimates for the labor force as a whole and for the self-employed are too imprecise to support any conclusion.

8 How Much of the Earnings Gain Do Foundational Skills Explain?

The results in Section 6 establish two reduced-form effects of DPEP eligibility for the same birth cohorts at the same threshold. As children, these cohorts acquired stronger foundational skills; as young adults in regular wage employment, they earned more. This section examines how much of the second effect can be accounted for by the first. I approach this question from two directions. Subsection 8.1 scales the earnings effect by the foundational skills effect. Because this ratio attributes the entire earnings gain to measured learning, it represents an upper bound on the learning channel. Subsection 8.2 constructs the opposite benchmark: the earnings effect DPEP would have produced if it had operated only through measured learning, valued using an observational skill-earnings relationship. Throughout, the earnings outcome is log weekly real earnings among regular wage and salaried employees, and all estimates use the MSE-optimal bandwidth and sampling weights unless otherwise noted.

8.1 An Upper Bound: The Earnings Effect per Unit of Foundational Learning

Because the threshold is common to both surveys, the ratio of the two discontinuities is a two-sample Wald estimator (Angrist and Krueger, 1992; Inoue and Solon, 2010),

$$\hat{\beta} = \frac{\hat{\tau}_Y}{\hat{\tau}_S}, \tag{3}$$

where $\hat{\tau}_Y$ is the discontinuity in log weekly earnings among regular employees shown in Table 5 and $\hat{\tau}_S$ is the discontinuity in the standardized composite test score presented in Table 4. If DPEP affected earnings only through measured foundational skills, $\hat{\beta}$ would identify the change in log earnings associated with a one-standard-deviation increase in early-grade learning among cohorts at the threshold. I report this ratio for all persons and for men.

Because $\hat{\beta}$ is a ratio of two estimates, I base inference on the method of Fieller (1954), which remains valid when the delta-method normal approximation to a ratio is inaccurate, as can occur when the denominator is only a few standard errors away from zero. The test-score discontinuity is significantly different from zero (Table 6, column 1), and the Fieller method directly incorporates this precision by testing restrictions that are linear in the two independent, approximately normal discontinuity estimates. For each candidate value β_0, I test $H_0 \colon \tau_Y - \beta_0 \tau_S = 0$ using

$$t(\beta_0) \;=\; \frac{\hat{\tau}_Y - \beta_0\,\hat{\tau}_S}{\sqrt{\mathrm{SE}(\hat{\tau}_Y)^2 + \beta_0^2\,\mathrm{SE}(\hat{\tau}_S)^2}}, \tag{4}$$

which treats the ASER and PLFS estimates as independent because they are based on independent samples. The 95 percent confidence set includes all values of β_0 for which $|t(\beta_0)| \leq 1.96$. At $\beta_0 = 0$, the statistic reduces to the t-statistic of the earnings discontinuity, so the test that the scaled effect is zero coincides with the test of $\hat{\tau}_Y$.

Table 6 reports the estimates. With the MSE-optimal bandwidths, the ratio is 2.05 log points per standard deviation for all persons and 1.76 for men. Both estimates are significantly different from zero at the 5 percent level. With the CER-optimal bandwidths, the ratios are 1.70 for all persons, significant at the 5 percent level, and 1.40 for men, significant at the 10 percent level. Expressed per 0.1 standard deviation, which corresponds to the scale of the learning gain DPEP actually produced, these ratios imply earnings 15 to 23 percent higher. With the MSE-optimal bandwidth, the 95 percent Fieller confidence sets range from 0.40 to 50.1 for all persons and from 0.28 to 26.3 for men; they are even wider with the CER-optimal bandwidth. The data therefore establish that the scaled effect is positive but offer little information about its magnitude.[11]

The point estimates are far larger than any plausible return to foundational skills alone. A coefficient of 2.05 implies that a one-standard-deviation improvement in early-grade learning would raise adult earnings nearly eightfold. By comparison, the Young Lives gradient between age-8 reading and age-29 earnings is 0.368 in Table 1. Observational estimates of the return to a standard deviation of measured cognitive skill are typically well below 0.5 (Hanushek et al., 2015; Aslam et al., 2012). The magnitude of $\hat{\beta}$ itself suggests that the exclusion restriction underlying Equation 3 fails. DPEP built schools, hired teachers, and supplied materials, so it could raise earnings through additional years of schooling, access to regular employment, and local labor markets, as well as through foundational learning. The ratio attributes all these channels to measured learning. I therefore interpret $\hat{\beta}$ as the earnings effect of the entire program expressed per standard deviation of learning. It is an upper bound on the return to foundational skills rather than an estimate of that return.

Whether DPEP also increased years of schooling matters directly for this interpretation because additional schooling is the most obvious violation of the exclusion restriction. The evidence on this question is less settled than is often assumed. Khanna (2023a) uses the same eligibility threshold and finds that DPEP increased completed schooling and wages. Roodman (2023) reanalyzes that study and reports that four districts close to the threshold that received DPEP are missing from the data. Restoring these districts reduces the estimated effect on schooling by about two-thirds and the effect on wages by about one-third. Roodman further shows that clustering standard errors by district, the unit at which eligibility is assigned, roughly doubles or triples them, leaving the estimates within one standard error of zero. Khanna (2023b) responds to Roodman (2023)'s reconstruction of the district sample and argues that the original results remain robust in Roodman (2023)'s sample.

[11] Appendix C estimates the same ratio in a single fuzzy RD within the PLFS sample, instrumenting the district-average ASER score with eligibility. Because that score is standardized across districts rather than across children, its coefficient (0.424 for all persons) is expressed per standard deviation of the district average; converted to individual standard deviations, it is close to the two-sample ratio reported here.

My estimates follow the practices at issue in this exchange. All standard errors are clustered at the district level. Bandwidths are selected separately for each outcome using the 'rdrobust' Stata package. Subsection 7.3 incorporates the four covered districts above the cutoff through a fuzzy design. Under these choices, the effects on test scores and on the earnings of regular employees remain statistically significant. Because the evidence on schooling remains contested, however, I cannot determine the direction of the bias in $\hat{\beta}$ by relying on prior estimates of DPEP's effect on educational attainment.

8.2 A Surrogate-Index Benchmark

The ratio in Subsection 8.1 attributes the entire earnings gain to foundational skills. I now construct a complementary benchmark using the surrogate-index framework of Athey et al. (2019): the earnings effect that DPEP eligibility would have produced if it had operated only through measured foundational skills. I compare this benchmark with the directly estimated PLFS earnings effect for regular wage and salaried employees, both for all persons and for men, and examine its sensitivity to the assumed skill-earnings gradient.

Athey et al. (2019) address a common problem in program evaluation: the outcome of interest, such as lifetime earnings, is observed only long after treatment, whereas intermediate outcomes, such as early test scores, are observed sooner. Their solution combines two samples. In an experimental sample, treatment and short-run outcomes are observed, but the long-run outcome is not. In an observational sample, the surrogate measures and the long-run outcome are observed together, but there is no exogenous variation in treatment. The relationship between the long-run outcome and the surrogates is estimated in the observational sample. The resulting prediction, known as the surrogate index, is then evaluated in the experimental sample. The treatment effect on the index equals the treatment effect on the long-run outcome under two assumptions. The first is surrogacy: conditional on the surrogates, treatment has no further effect on the long-run outcome. The second is comparability: the conditional relationship between the long-run outcome and the surrogates is the same in both samples.

My setting maps onto this framework. The treatment is DPEP eligibility, assigned according to the 1991 female literacy threshold. The experimental sample is ASER, in which I estimate the discontinuity in foundational skills at the threshold. The observational sample is the YLS, which follows children from age 8 to age 29 and records both early reading and adult earnings for the same individuals. The long-run outcome is log earnings. Unlike most applications of the surrogate-index framework, I also observe the long-run outcome directly through the earnings discontinuity in the PLFS. I therefore use the index as a diagnostic rather than as a substitute for direct estimation. If DPEP raised earnings only through foundational skills, and if the YLS skill-earnings gradient applies to districts near the threshold, the predicted and observed effects should coincide. A gap indicates either that DPEP raised earnings through channels other than measured foundational skills or that the skill-earnings relationship differs between the YLS sample and the districts at the cutoff.

Let S denote foundational skill in standard deviation units and Y log earnings. From the observational sample, I take the gradient γ of log weekly labor income at age 29 on the standardized age-8 reading score, estimated in Equation 1; from Table 1, $\hat{\gamma} = 0.368$. The surrogate index for an individual is $\hat{\gamma}S$. Let $\hat{\tau}_S$ denote the RD effect of DPEP eligibility on

standardized test scores in ASER (Table 4). Because the index is linear in a single surrogate, the RD effect of eligibility on the index is

$$\hat{\tau}_Y^{\mathrm{SI}} = \hat{\gamma} \times \hat{\tau}_S. \tag{5}$$

Under surrogacy and comparability, $\hat{\tau}_Y^{\mathrm{SI}}$ estimates the effect of DPEP eligibility on log earnings. Because the YLS and ASER samples are independent, the delta-method standard error is

$$\mathrm{SE}\big(\hat{\tau}_Y^{\mathrm{SI}}\big) = \sqrt{\hat{\gamma}^2\,\mathrm{SE}(\hat{\tau}_S)^2 + \hat{\tau}_S^2\,\mathrm{SE}(\hat{\gamma})^2} \tag{6}$$

I compare this prediction with the directly estimated effect $\hat{\tau}_Y$ (Table 5). I report the difference $\hat{\tau}_Y - \hat{\tau}_Y^{\mathrm{SI}}$, whose standard error combines the two independent standard errors, and the share of the observed effect accounted for by the index, $\hat{\tau}_Y^{\mathrm{SI}}/\hat{\tau}_Y$. This exercise requires the skill measures in the two samples to be on a common scale. I assume that one standard deviation of the YLS age-8 reading score corresponds to one standard deviation of the ASER test score, since both are standardized measures of early-grade learning within their respective samples. For men, I use the gradient estimated for the male YLS sample, $\hat{\gamma}_m = 0.387$ (Table A1). I do not construct a benchmark for women because both the female YLS gradient and the PLFS earnings effect for women are statistically insignificant.

Table 7 reports the results. DPEP eligibility raises standardized test scores by 0.137 standard deviations. Combined with the YLS gradient of 0.368, this implies a surrogate-index prediction of 0.050 log points, or about 5 percent higher earnings. This is the gain one would expect at the threshold if DPEP had raised earnings only by improving foundational skills and if the YLS relationship between early reading and adult income applied to these districts. The directly estimated effect for regular employees, 0.281 log points, is more than five times as large, so the index accounts for about 18 percent of the observed effect. The difference of 0.231 log points is significant at the 5 percent level. Valued at the YLS gradient, foundational skills therefore explain approximately one-fifth of the earnings gain.

The pattern is similar for men. The male test-score discontinuity of 0.141 standard deviations, combined with the male YLS gradient of 0.387, implies a predicted earnings effect of 0.055 log points. The observed effect for male regular employees is 0.248, so the index accounts for about 22 percent of it. The difference of 0.194 log points is significant at the 10 percent level. The male and pooled results are close because both the test-score effect and the skill-earnings gradient for men are only slightly larger than their pooled counterparts.

How plausible is the YLS gradient as a measure of the earnings value of foundational skills in developing countries? A long line of studies in low- and middle-income countries finds that directly tested literacy and numeracy skills are rewarded in the labor market beyond the returns to years of schooling. In Kenya and Tanzania, wage employees with higher reading and numeracy scores earned substantially more, and much of the apparent return to schooling operated through these skills (Boissiere et al., 1985). In Ghana, cognitive skills were associated with higher household income, particularly from off-farm activities (Jolliffe, 1998). In rural Pakistan, returns to school quality, measured by cognitive achievement, were comparable to or greater than returns to additional years of schooling (Behrman et al., 2008), and literacy and numeracy scores predicted wages in wage employment (Aslam et al., 2012). Glewwe (2002) reviews this literature and concludes that basic cognitive skills, rather than

time spent in school, account for much of the return to education in developing countries. More recent evidence from Indonesia links foundational literacy and numeracy directly to adult earnings (Crawfurd, 2026). Two features of this literature matter for my benchmark. First, it measures adults' skills at the time of the survey rather than skills acquired in early childhood, and its estimates are correlational. Second, it suggests that returns to foundational skills are concentrated in wage employment, where reading and arithmetic are used directly and screened for by employers. This pattern is consistent with my finding that the earnings effects of DPEP appear among regular employees but not among the self-employed. For India, the evidence is thinner still because few surveys follow children into adulthood; this is why I rely on the YLS for the gradient.

Experimental evidence from India and other developing countries is informative about the size of the learning effect, though not yet about its earnings value. Randomized evaluations of remedial instruction, teacher incentives, teaching at the right level, and technology-aided learning in India have produced gains in foundational learning broadly similar in magnitude to the effect I estimate (Banerjee et al., 2007; Muralidharan and Sundararaman, 2011; Duflo et al., 2012; Banerjee et al., 2017; Muralidharan et al., 2019). Across education interventions evaluated in low- and middle-income countries, the median effect on learning is about 0.1 standard deviations (Evans and Yuan, 2022). The DPEP effect of 0.137 standard deviations is therefore typical of successful interventions, so the gap between predicted and observed earnings effects does not stem from an unusually small learning effect. Few of these trials have followed participants into the labor market (Bouguen et al., 2019). Instead, the earnings value of learning gains in developing countries is usually projected by applying an observational skill-earnings gradient to experimental test-score effects (Angrist et al., 2025), which is the logic of the surrogate index. Evidence from high-income settings shows that such projections can understate long-run effects because programs also build skills that test scores do not capture (Chetty et al., 2011; Heckman et al., 2013; Jackson, 2018). My setting is unusual in that I observe both the learning effect and the earnings effect of the same program for the same cohorts. The comparison in Table 7 suggests that, for a program that expanded access as well as learning, projections based on learning gains alone would capture only a small fraction of the earnings effect.

Table 8 shows how the benchmark varies with the assumed gradient. Rescaling the gradient rescales its standard error by the same factor, so the precision of the prediction relative to its size remains unchanged across rows. Suppose the YLS gradient overstates the causal return to early reading by a factor of two, for example, because age-8 reading partly reflects unobserved family investments. Measured learning then accounts for about 9 percent of the earnings effect for all persons and 11 percent for men. At the YLS gradients themselves, the shares are 18 and 22 percent. If instead the YLS reading score measures early skill with a reliability of only one-half, so that the gradient is attenuated by a factor of two, the shares rise to 36 and 44 percent. A more direct way to interpret the evidence is to ask what gradient would be needed for learning to explain the entire earnings effect. This break-even gradient, $\gamma^* = \hat{\tau}_Y/\hat{\tau}_S$, is the ratio $\hat{\beta}$ of Subsection 8.1: 2.05 log points per standard deviation for all persons and 1.76 for men. It implies that a one-standard-deviation improvement in early-grade learning would raise adult earnings six- to eightfold, far beyond what the developing-country evidence on returns to cognitive skills suggests.

Several considerations qualify this interpretation, and they work in different directions.

Surrogacy is unlikely to hold because DPEP may have raised years of schooling as well as learning; the index is therefore a benchmark for the learning channel rather than a full prediction of the earnings effect. Comparability requires the YLS gradient to apply to districts near the DPEP threshold. The YLS covers purposively selected sites in Andhra Pradesh and Telangana and is not representative of India. A standard deviation of YLS reading need not correspond to a standard deviation of the ASER composite, which combines reading with arithmetic. The YLS gradient is estimated at age 29 for all earners, whereas the PLFS outcome is the earnings of regular employees aged 22-25. The YLS estimates rest on 633 observations in 20 sites (381 for men) and remain correlational. Because returns to foundational skills in developing countries appear concentrated in wage employment, applying a gradient estimated on all earners to regular employees may understate the relevant return. Finally, the earnings effect is estimated conditional on being a regular employee, so any change in who holds these jobs affects the observed effect but not the index. For these reasons, I present the comparison as a diagnostic benchmark rather than as a decomposition.

The two exercises bracket the role of foundational learning. The ratio in Subsection 8.1 assigns the entire earnings effect to learning and implies an implausibly large return. The surrogate index in Subsection 8.2 values the learning gain at observational gradients. Under plausible gradients, it accounts for between about one-tenth and just over one-third of the effect for all persons, and up to about 44 percent for men. Together, the two benchmarks indicate that improved foundational learning is an important channel through which DPEP raised earnings. Natural candidates for the remainder are additional schooling, changes in access to regular employment, skills that test scores do not capture, and any local labor-market effects of the program.

9 Conclusion and Discussion

This paper asks whether a program that improved foundational learning in primary school also improved labor market outcomes in early adulthood in India. No nationally representative Indian dataset follows children into the labor market, so I link two independent surveys that observe the same birth cohorts at different stages of life. The Annual Status of Education Report measures the reading and arithmetic of rural children aged 8–11 between 2006 and 2011, and the Periodic Labour Force Survey records the employment and earnings of rural young adults aged 22–25 between 2020 and 2025. The eligibility rule of the District Primary Education Programme, which targeted districts whose 1991 female literacy rate fell below the national average, generates a discontinuity common to both surveys. I exploit this discontinuity in a regression discontinuity design to estimate the program's effects at both stages of life.

The evidence supports three conclusions. First, DPEP eligibility raised foundational learning. Children in districts just eligible for the program score 0.137 standard deviations higher on a composite measure of the ability to read a Grade 2 story and to solve a division problem, an increase of about 19 percent relative to the sample mean. The effect is nearly identical for boys and girls, and it holds across bandwidth selectors, with and without sampling weights, under linear and quadratic specifications, and when eligibility is replaced by actual program coverage, with estimates ranging from 0.14 to 0.25 standard deviations. Sec-

ond, the same cohorts earn more in the best-paid segment of the labor market. Among young adults in regular wage or salaried employment, weekly earnings are about 32 percent higher in districts just eligible for DPEP, and the gain remains positive in every specification, at between roughly 15 and 41 percent. It is precisely estimated for men and at least as large but imprecise for women. I find no reliable effect among the self-employed or in the labor force as a whole, and the negative estimates for casual laborers disappear once the sampling weights are removed. Third, measured learning accounts for only part of the earnings gain. Scaling the earnings effect by the learning effect yields about 2 log points per standard deviation of learning, far more than any plausible return to foundational skills alone. A surrogate-index benchmark that values the learning gain at the Young Lives skill-earnings gradient accounts for about one-fifth of the observed effect, and for between roughly one-tenth and just over one-third under plausible alternative gradients.

These estimates should be read as the effects of a program that demonstrably raised foundational learning, not as the return to a unit of foundational skill. DPEP was a bundled intervention. It built schools, hired teachers, and supplied learning materials, and earlier work finds that it also raised enrollment and completed years of schooling (Azam and Saing, 2017; Khanna, 2023a), although the size of these effects remains contested (**?**Khanna, 2023b). The earnings gains I document therefore combine the effect of stronger foundational skills with the effects of additional schooling and of any other channel through which the program operated. The two benchmarks in Section 8 bracket the share attributable to measured learning: it is clearly above zero, but well below one. The same comparison carries a methodological implication. The earnings value of education interventions in developing countries is usually projected by applying an observational skill-earnings gradient to test-score gains. For a program that expanded access as well as learning, such a projection would have captured only a fraction of the earnings effect observed directly.

The concentration of the earnings gains in regular wage and salaried employment is nonetheless informative. Regular jobs are more stable, better paid, and more likely to carry benefits than casual labor or self-employment, and they are the jobs in which literacy and numeracy are most plausibly used and rewarded: in following written instructions, keeping records, handling transactions, and meeting the screening requirements of formal employers. That eligibility raises earnings within this segment, while effects elsewhere are absent or fragile, is consistent with foundational skills acting both as a gateway to better-quality wage employment and as a source of productivity within it. It also mirrors the developing-country evidence that returns to tested literacy and numeracy are concentrated in wage work. Because the employment-type estimates condition on the kind of job held, however, they combine any rise in productivity within regular jobs with any change in who holds them, and my earnings estimates alone cannot establish whether DPEP widened access to regular employment.

The results for women require particular care. DPEP selected districts on the basis of female literacy and explicitly sought to narrow gender gaps in schooling, and at the threshold its effect on foundational learning was shared almost equally by boys and girls. Whether these gains translated into labor market returns for women is much harder to establish. Women make up only one in four members of my labor-force sample, and fewer than 3,000 hold regular employment. For women in regular jobs, the estimated earnings gain is as large as or larger than that for men, but it is imprecise and sensitive to the bandwidth.

Earnings are observed only for the minority of young women in the labor force, and if DPEP affected whether women enter paid work, the estimates condition on a selected group. The implication is not that foundational skills matter less for women, but that whether improved learning appears in measured earnings depends on access to the labor market itself. In India, that access remains constrained by low female labor force participation, social norms, and household responsibilities (Klasen and Pieters, 2015; **?**; Jayachandran, 2021). My design cannot speak to this margin, which is an important question in its own right.

The findings bear directly on a defining feature of Indian education over the past three decades: access has expanded far faster than learning. Primary enrollment is now nearly universal, yet ASER has documented for two decades that a large share of children in the primary and upper-primary grades cannot read a simple text or perform basic arithmetic. In my own sample, fewer than four in ten children aged 8–11 could read a Grade 2 story, and the share who could perform division fell from 30 percent in 2006 to 19 percent in 2011. My results show that a large-scale public program moved these outcomes and that the cohorts it reached earn more as young adults in regular wage employment. This provides quasi-experimental evidence for an argument prominent in the development literature, that in many low- and middle-income countries the binding constraint is the quality of learning rather than the quantity of schooling. It also aligns with the National Education Policy 2020, which places universal foundational literacy and numeracy at the center of school reform (**?**). My estimates suggest that this objective carries an economic payoff, and the surrogate-index comparison suggests that valuing such programs by their test-score gains alone may understate it.

Several limitations qualify these conclusions. The earnings estimates capture the combined effect of the program rather than the isolated return to foundational skills, and the surrogate benchmark rests on a correlational gradient from a purposively sampled, non-representative study in a single region. The estimates are local: they identify effects for districts near the eligibility threshold, which had low female literacy by national standards in 1991. This is the margin most relevant for targeted intervention, but the effects need not generalize to districts far from the cutoff. The ASER and PLFS samples cover the same birth cohorts in the same districts but not the same individuals, and both record district of residence rather than district of schooling. Because internal migration in India is limited and mostly short-distance (Munshi and Rosenzweig, 2016), the resulting misclassification is likely to attenuate rather than inflate my estimates. I observe these cohorts only at ages 22–25, at the start of their working lives, when some are still in education and many have yet to settle into stable employment; if returns to skill grow with experience, the long-run gains are likely to be larger. Finally, my comparison of districts on either side of the threshold does not capture general-equilibrium effects. By raising the supply of better-educated workers, DPEP may have changed skill prices in local labor markets (Khanna, 2023a), and such effects would also reach workers whose own skills the program did not improve.

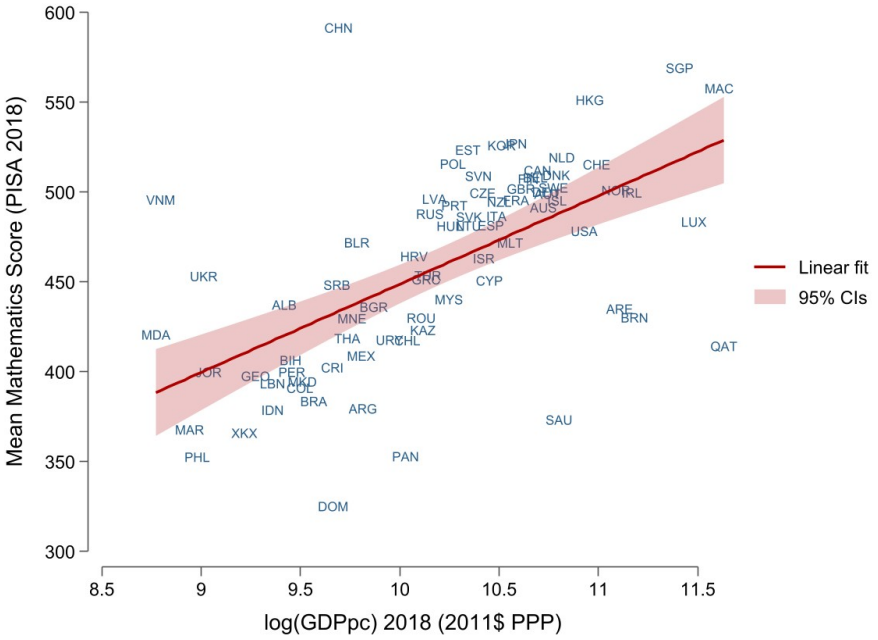
These limitations point to natural extensions. Older cohorts in the PLFS, who had already passed primary-school age when DPEP began, provide a placebo: any earnings discontinuity among them would reflect changes in local labor markets rather than in their own skills, and comparing older and younger cohorts at the threshold would help separate the direct effect of improved learning from general-equilibrium effects. Estimating the effect of eligibility on labor force participation, on the probability of holding each type of job, and on

completed schooling would show whether the program widened access to regular employment and how much of the earnings gain operates through additional years of schooling. Following these cohorts in future PLFS rounds would reveal whether the earnings gains widen, persist, or fade over the life cycle.

More broadly, the paper shows how the long-run consequences of early learning can be studied where the ideal longitudinal data do not exist. Few developing countries track the same children from primary school into the labor market. Linking independent surveys of learning and of employment through birth cohort and place, and pairing them with a policy rule that shifts learning for some cohorts and not others, offers a template for other data-scarce settings, and it allows the projections commonly used to value education interventions to be checked against observed long-run effects. In a country where schooling has expanded without commensurate learning, a program that improved foundational skills in childhood was followed by higher earnings in early adulthood among those who reached the better-paid segment of the labor market. Whether such gains scale to universal programs, extend to women's participation in paid work, and persist over the life cycle are questions these local estimates can motivate but not settle.

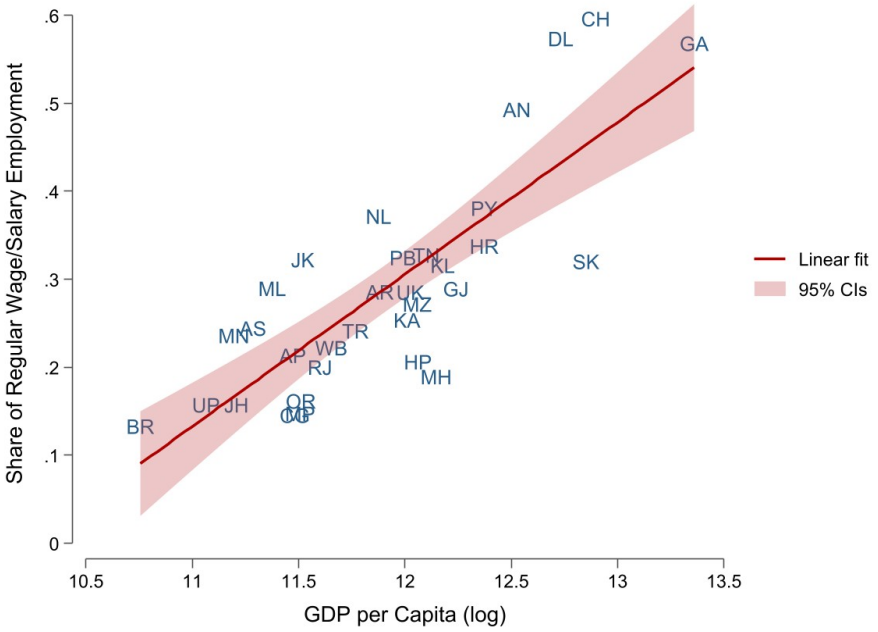
Figures and Tables

Figure 1: Cross-Country Correlation Between GDP and PISA Scores



Note: The figure plots mean mathematics scores from PISA 2018 and log GDP per capita 2018 in constant 2011 PPP-adjusted US dollars for a cross-section of countries. The red line is the OLS linear fit with the 95% confidence interval. Both variables are sourced from World Bank databases: PISA scores from Education Statistics and GDP per capita from World Development Indicators.

Figure 2: Cross-State Correlation Between GDP and Regular Wage/Salary Employment



Note: The figure plots the share of regular wage/salaried employment and log GDP per capita at constant prices across Indian states in 2018. Each point represents a state. The red line is the OLS linear fit with the 95% confidence interval. Employment shares are computed for employed individuals aged 21-60 using the Periodic Labour Force Survey (PLFS) 2017-18, sourced from the Ministry of Statistics and Programme Implementation (MoSPI). State GDP per capita is sourced from MoSPI.

Figure 3: Cohort Timeline and Survey Coverage

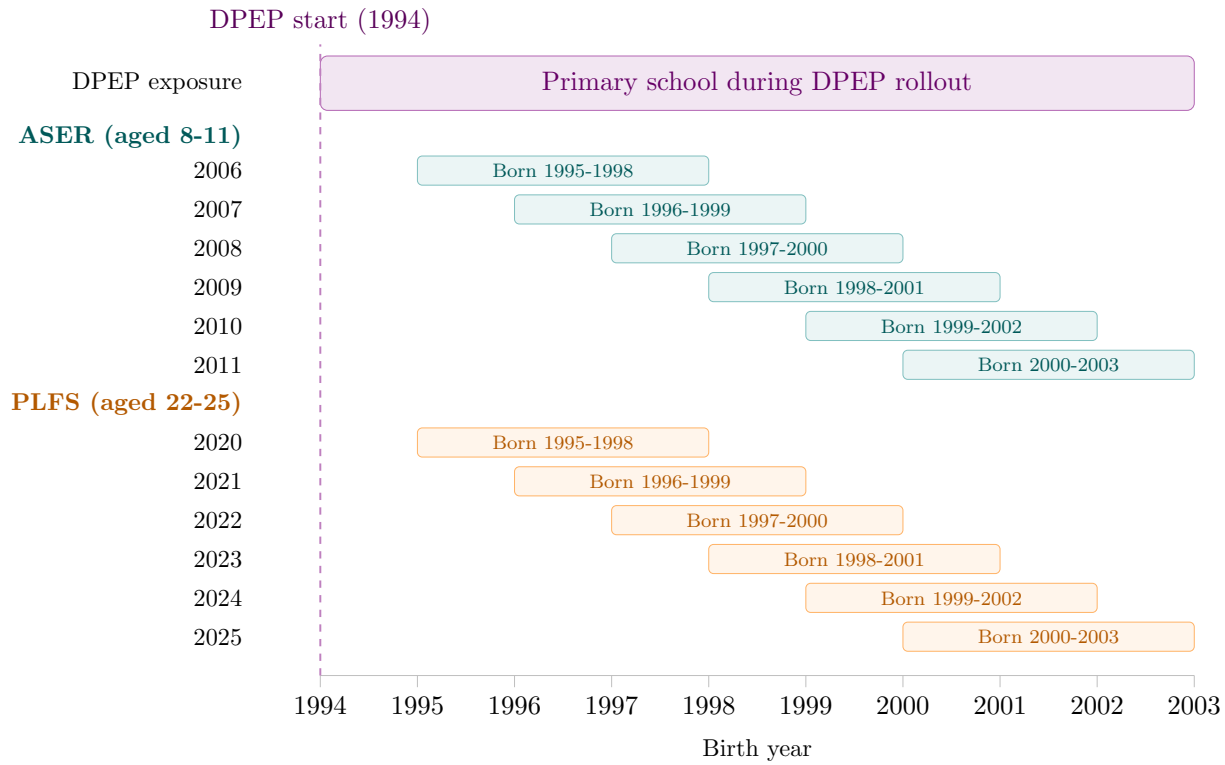
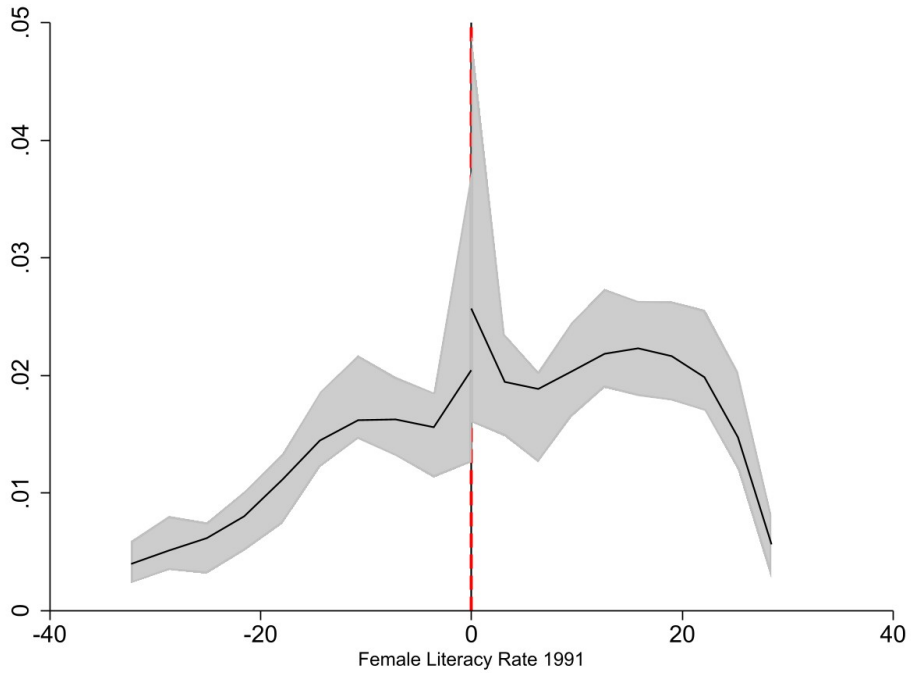


Figure 4: Continuity of the Running Variable



Notes: 'Female Literacy Rate 1991' is the negative of the district-level female literacy rate from the 1991 Census, centered at the cutoff. The figure plots the estimated density of the running variable on either side of the cutoff using local polynomial density estimation with a triangular kernel and combined bandwidth selector. The robust T-statistic of 0.701 (p-value = 0.484) fails to reject the null hypothesis of no manipulation in the density of the running variable at the cutoff.

Table 1: OLS Estimates: Foundational Skills and Later-Life Outcomes (YLS)

	(1)	(2)	(3)	(4)	(5)
	Cognitive Skills		Non-Cognitive Skills		Labor Market Outcomes
	CLOZE Score (15 yrs)	PPVT Score (15 yrs)	Self-Esteem Score (22 yrs)	Socio-Emotional Score (22 yrs)	Log Weekly Income (29 yrs)
Reading Score (8 yrs)	0.356***	0.296***	-0.005	0.086***	0.368***
	(0.038)	(0.038)	(0.044)	(0.030)	(0.122)
Female	-0.305***	-0.380***	-0.130	-0.411***	-1.956***
	(0.069)	(0.076)	(0.085)	(0.085)	(0.246)
Child's Age (months)	0.005	-0.009	0.012	0.012	-0.002
	(0.009)	(0.010)	(0.007)	(0.011)	(0.028)
Household Size	-0.004	0.007	0.003	0.017	-0.088
	(0.013)	(0.012)	(0.014)	(0.018)	(0.084)
Scheduled Tribe (ST)	-0.220	-0.119	0.145	0.312***	-1.290**
	(0.162)	(0.118)	(0.099)	(0.088)	(0.565)
Backward Class (BC)	-0.046	0.040	0.158	0.107	-0.206
	(0.075)	(0.084)	(0.108)	(0.077)	(0.310)
Other Castes (OC)	0.120	0.241*	0.484***	0.292**	-0.188
	(0.124)	(0.122)	(0.097)	(0.123)	(0.335)
Muslim	-0.449*	-0.348**	-0.466**	-0.159	0.263
	(0.253)	(0.166)	(0.223)	(0.219)	(0.711)
Buddhist	-0.067	-0.079	0.351	0.342	1.227*
	(0.252)	(0.194)	(0.205)	(0.237)	(0.632)
Hindu	0.107	-0.067	0.069	0.211	-0.469
	(0.106)	(0.095)	(0.178)	(0.170)	(0.460)
Sikh	0.498***	0.505***	2.186***	-	2.249***
	(0.155)	(0.113)	(0.155)		(0.517)
Caregiver Did Not Complete Primary	-0.286***	-0.253***	-0.073	-0.252**	-0.120
	(0.076)	(0.073)	(0.120)	(0.111)	(0.289)
Wealth Index	0.944***	0.990***	0.030	0.341	0.639
	(0.187)	(0.188)	(0.229)	(0.230)	(0.586)
Cluster FE	Yes	Yes	Yes	Yes	Yes
SE Clustered at	Cluster	Cluster	Cluster	Cluster	Cluster
Number of Clusters	20	20	20	20	20
Observations	885	935	839	772	633
R^2	0.300	0.339	0.085	0.220	0.196

Notes: Standard errors clustered at the sentinel-site (cluster) level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All specifications include cluster fixed effects. The omitted categories are male (gender), Scheduled Caste (caste), Christian (religion), and caregiver completed primary schooling or more (caregiver education). Controls include the child's age in months, household size, caste, religion, caregiver education, and the household wealth index. - indicates the category is omitted because no observations fall in it in that sample. The sample comprises children who were 8 years old in the first wave of the Young Lives Survey in India in 2002 and were subsequently followed at ages 15 (2009), 22 (2016), and 29 (2023-24). All scores are standardized. The Reading Score measures foundational literacy at age 8 (Round 1). The CLOZE Score is a sentence-completion test that measures reading comprehension at age 15 (Round 3). The PPVT Score, based on the Peabody Picture Vocabulary Test, measures receptive vocabulary at age 15 (Round 3). The Self-Esteem Score is a polychoric PCA index constructed from 31 self-perception items measured on a 1-4 scale at age 22 (Round 5). The Socio-Emotional Score is a polychoric PCA index constructed from 26 psychosocial and personality items measured on a 1-5 scale at age 22 (Round 5). Earnings are measured as $\ln(1 + \text{weekly labor income})$, where income combines cash and in-kind payments converted to weekly equivalents, at age 29 (Round 7).

Table 2: Summary Statistics: ASER Test Scores

	Observations	Mean	Std. Dev.	Min	Max
All children	1,298,036	0.605	0.821	0	2
Boys	703,826	0.616	0.829	0	2
Girls	594,210	0.592	0.812	0	2

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The test score is constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively, and ranges from 0 to 2. In the regressions, the dependent variable is this score standardized over the pooled sample.

Table 3: Summary Statistics: PLFS Weekly Real Earnings

	Observations	Mean	Std. Dev.	Min	Max
Panel A: All persons					
All workers	79,548	776.58	884.44	0.00	6127.66
Self-employed	35,791	548.03	836.99	0.00	6127.66
Regular salaried	15,277	1576.65	876.53	0.00	6127.66
Casual labourers	15,681	1152.63	494.20	0.00	4622.69
Panel B: Males					
All workers	59,551	907.97	910.45	0.00	6127.66
Self-employed	23,895	711.31	945.38	0.00	6127.66
Regular salaried	12,550	1620.09	831.89	0.00	6127.66
Casual labourers	13,940	1200.97	485.16	0.00	4622.69
Panel C: Females					
All workers	19,993	385.34	662.38	0.00	6127.66
Self-employed	11,892	220.05	389.12	0.00	5830.42
Regular salaried	2,727	1376.75	1034.89	0.00	6127.66
Casual labourers	1,741	765.54	382.87	85.11	2798.60

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. Weekly real earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices, and winsorized at the 1st and 99th percentiles by year. Rows are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). In the regressions, the dependent variable is the log of weekly real earnings.

Table 4: RD Estimates of Effect on Standardized Test Scores

	All	Boys	Girls
	(1)	(2)	(3)
Panel A: MSE-optimal bandwidth			
RD estimate	0.137**	0.141**	0.133*
	(0.067)	(0.067)	(0.068)
Bandwidth (h)	11.722	11.722	11.722
Bias bandwidth (b)	18.554	18.554	18.554
Effective observations	550,344	296,493	253,851
Observations	1,298,036	703,826	594,210
Clusters (districts)	399	399	399
Panel B: CER-optimal bandwidth			
RD estimate	0.152**	0.159**	0.144*
	(0.077)	(0.077)	(0.078)
Bandwidth (h)	8.543	8.543	8.543
Bias bandwidth (b)	18.554	18.554	18.554
Effective observations	364,793	195,894	168,899
Observations	1,298,036	703,826	594,210
Clusters (districts)	399	399	399

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The dependent variable is the standardized test score, constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively. Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust using ASER sampling weights and adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: RD Estimates of Effect on Log Weekly Real Earnings

	All persons				Males				Females			
	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: MSE-optimal bandwidth												
RD estimate	-0.172	-0.226	0.281**	-0.160**	-0.116	-0.512	0.248**	-0.105	-0.515	0.139	0.336*	-0.334*
	(0.277)	(0.525)	(0.112)	(0.081)	(0.304)	(0.545)	(0.105)	(0.067)	(0.466)	(0.716)	(0.191)	(0.179)
Bandwidth (h)	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459
Bias bandwidth (b)	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259
Effective observations	24,805	10,204	5,289	5,221	18,858	7,108	4,326	4,499	5,945	3,094	963	722
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	313	313	313	307	313	313	312	307	313	308	260	211
Panel B: CER-optimal bandwidth												
RD estimate	-0.267	-0.428	0.259**	-0.211**	-0.187	-0.716	0.223*	-0.141*	-0.701	-0.124	0.305	-0.406**
	(0.309)	(0.614)	(0.128)	(0.096)	(0.342)	(0.622)	(0.117)	(0.075)	(0.537)	(0.835)	(0.209)	(0.203)
Bandwidth (h)	6.888	6.888	6.888	6.888	6.888	6.888	6.888	6.888	6.888	6.888	6.888	6.888
Bias bandwidth (b)	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259
Effective observations	18,967	7,671	4,203	3,949	14,381	5,290	3,425	3,446	4,584	2,379	778	503
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	313	313	313	307	313	313	312	307	313	308	260	211

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. The dependent variable is the log of weekly real earnings, where weekly earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices. Columns are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust using PLFS sampling weights and adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Earnings Effect per Standard Deviation of Learning

	(1) Effect on test score (SD)	(2) Effect on log earnings	(3) Ratio (2)/(1)	(4) Earnings gain per 0.1 SD (%)
Panel A: MSE-optimal bandwidth				
All persons	0.137**	0.281**	2.051**	22.8
	(0.067)	(0.112)	[0.40, 50.1]	
Men	0.141**	0.248**	1.759**	19.2
	(0.067)	(0.105)	[0.28, 26.3]	
Panel B: CER-optimal bandwidth				
All persons	0.152**	0.259**	1.704**	18.6
	(0.077)	(0.128)	[0.05, 240.7]	
Men	0.159**	0.223*	1.403*	15.1
	(0.077)	(0.117)	[−0.04, 28.4]	

Notes: Column (1) reports the sharp RD effect of DPEP eligibility on the standardized ASER composite test score for children aged 8-11. Column (2) reports the sharp RD effect on the log of one plus weekly real earnings for regular wage and salaried employees aged 22-25 in the PLFS. Both use local linear regression with a triangular kernel, sampling weights, the mass-point adjustment, and district-clustered standard errors (in parentheses). Column (3) is the ratio of column (2) to column (1), a two-sample Wald estimate. Inference on the ratio uses the Fieller method, treating the ASER and PLFS estimates as independent; brackets report the 95 percent Fieller confidence set. Column (4) reports the percentage earnings difference implied by a 0.1 standard deviation difference in learning. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Surrogate-Index Benchmark for the Earnings Effect of DPEP Eligibility

	(1) Effect on test score (SD)	(2) Surrogate-predicted earnings effect	(3) Observed effect, regular employees	(4) Difference (3) − (2)	(5) Share explained (2)/(3)
All persons	0.137**	0.050*	0.281**	0.231**	0.18
	(0.067)	(0.030)	(0.112)	(0.116)	
Men	0.141**	0.055*	0.248**	0.194*	0.22
	(0.067)	(0.032)	(0.105)	(0.110)	

Notes: All estimates use the MSE-optimal bandwidth and sampling weights. Column (1) reports the RD effect of DPEP eligibility on the standardized ASER composite test score. Column (2) multiplies column (1) by the YLS coefficient of log weekly labor income at age 29 on the standardized age-8 reading score. Column (3) reports the RD effect on log weekly real earnings for regular wage and salaried employees in the PLFS. Column (5) is the ratio of column (2) to column (3); inference on the gap between predicted and observed effects is given by column (4). Standard errors in parentheses; those in columns (2) and (4) are computed by the delta method, treating the YLS, ASER, and PLFS estimates as independent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Sensitivity of the Surrogate-Index Benchmark to the Skill-Earnings Gradient

	All persons			Men		
	Gradient γ (1)	Predicted effect (2)	Share explained (3)	Gradient γ_m (4)	Predicted effect (5)	Share explained (6)
Panel A: Predicted earnings effect under alternative gradients						
YLS gradient halved	0.184***	0.025*	0.09	0.194***	0.027*	0.11
	(0.061)	(0.015)		(0.068)	(0.016)	
YLS gradient (baseline)	0.368***	0.050*	0.18	0.387***	0.055*	0.22
	(0.122)	(0.030)		(0.135)	(0.032)	
YLS, corrected for reliability of 0.5	0.736***	0.101*	0.36	0.774***	0.109*	0.44
	(0.244)	(0.060)		(0.270)	(0.064)	
Panel B: Gradient required to explain the full observed effect						
Break-even gradient, $\gamma^* = \hat{\tau}_Y/\hat{\tau}_S$	2.051**	0.281	1.00	1.759**	0.248	1.00
	[0.40, 50.1]			[0.28, 26.3]		

Notes: The predicted effect is $\gamma \times \hat{\tau}_S$, where $\hat{\tau}_S$ is the RD effect of DPEP eligibility on the standardized ASER composite test score. The share explained is the predicted effect divided by the observed RD effect on log weekly real earnings of regular employees. The baseline gradients are the YLS coefficients of log weekly labor income at age 29 on the standardized age-8 reading score, estimated on the pooled sample for all persons and on the male sample for men. The first row halves these gradients, allowing for upward bias from unobserved family investments; the third row divides them by an assumed reliability of 0.5, allowing for attenuation from measurement error. Both adjustments are illustrative and rescale the YLS estimate and its standard error by the same factor. In Panel A, standard errors are in parentheses; those for predicted effects are computed by the delta method, treating the YLS and ASER estimates as independent, and stars are based on the ratio of each estimate to its standard error. Panel B reports the gradient at which the surrogate index would account for the entire observed effect. The predicted effect and share in Panel B equal the observed effect and one by construction. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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Appendices

A Data Description

A.1 Annual Status of Education Report

ASER is a household-based survey of schooling and learning conducted annually since 2005 by the non-governmental organization Pratham, with the help of volunteers from local partner organizations. It covers nearly every rural district in India and is designed to produce estimates of schooling and learning that are representative at the district level. In each district, the survey samples villages using Census population data and then samples households within each village. All children aged 3-16 in sampled households are surveyed, and children aged 5-16 are assessed.

A defining feature of ASER is its direct, one-on-one assessment of foundational literacy and numeracy, conducted in the child's home rather than at school. Assessing children at home, regardless of their enrollment status or attendance on the day of the survey, avoids the selection associated with school-based tests and the reporting biases that arise when enrollment or grade attainment is used as a proxy for learning. The reading test is administered in the child's preferred language. Both tests follow a fixed, ordered sequence of tasks, and the assessor records only the highest level the child completes comfortably. The assessment tools remained unchanged across the rounds used in this study, making the recorded levels comparable across years.

Alongside the assessments, ASER records a limited set of child and household characteristics, including age, sex, enrollment status, grade, and school type. It does not follow the same children over time or collect information on earnings and other adult outcomes. I pool the rounds from 2006 to 2011 and restrict the sample to children aged 8-11 to measure foundational skills for the birth cohorts whose adult earnings I observe in the PLFS. The construction of the outcome variables is described in Section 3.

A.2 Periodic Labour Force Survey

The PLFS is conducted by the National Statistics Office under the Ministry of Statistics and Programme Implementation, Government of India. Introduced in 2017-18, it generates regular estimates of key labor market indicators, including labor force participation, the worker-population ratio, unemployment, and the distribution of workers across industries, occupations, and employment types. The survey uses a stratified multistage sampling design, with villages in rural areas and urban blocks in urban areas serving as first-stage sampling units and households as second-stage units. Each household is assigned a sampling multiplier, which I use to weight all estimates.

It records demographic characteristics, education, and activity status for every household member under two reference periods: usual status, which refers to the preceding 365 days, and current weekly status, which refers to the preceding seven days. Earnings are recorded in the current weekly status framework. For casual laborers, the survey records wage earnings for each day of the reference week, for up to two activities (principal and subsidiary) per

day. For regular wage and salaried employees, it records earnings for the preceding calendar month. For the self-employed, it records earnings from self-employment over the preceding 30 days. Unlike earlier employment surveys in India, which recorded earnings only for wage workers, the PLFS records earnings for all three categories of workers. This enables me to construct a comparable measure of weekly earnings for the entire employed population. The unit-level data also identify the district in which each household is located.

I pool the survey rounds from 2020 to 2025 and restrict the sample to rural individuals aged 22-25, corresponding to the birth cohorts observed at ages 8-11 in ASER. The construction of the earnings and employment variables is described in Section 3.

A.3 Young Lives Study

The YLS is a longitudinal study of childhood poverty coordinated by the Department of International Development at the University of Oxford. It follows children in four countries: Ethiopia, India, Peru, and Vietnam. In India, the study is conducted in Andhra Pradesh and Telangana, which formed the undivided state of Andhra Pradesh when the study began. It is one of the very few surveys in India that follow the same children from primary school into adulthood and record both their early learning and their later labor market outcomes.

The Indian sample was drawn in 2002 from 20 sentinel sites selected purposively to capture the diversity of the state across regions and rural and urban areas, with poorer areas over-represented. Within each site, households with children in the target age groups were selected at random. The study follows two cohorts: a younger cohort of approximately 2,000 children born in 2001-02 and an older cohort of approximately 1,000 children born in 1994-95. Both cohorts have been re-interviewed in successive rounds since 2002, with relatively low attrition by the standards of long-running panels in developing countries. Because the sites were not selected randomly, the sample is not representative of India or of either state. Its principal value lies in tracking the same individuals over two decades.

I use the older cohort, whose members were approximately 8 years old in Round 1 (2002), 15 in Round 3 (2009), 22 in Round 5 (2016), and 29 in Round 7 (2023-24). I measure foundational literacy using the reading assessment administered at age 8 and standardize the scores within the sample. At age 15, I measure cognitive skills using a sentence-completion (CLOZE) test of reading comprehension and the Peabody Picture Vocabulary Test (PPVT), which assesses receptive vocabulary. At age 22, the survey records a battery of self-reported psychosocial items, which I summarize in two indices. The self-esteem index is constructed through polychoric principal component analysis of 31 self-perception items measured on a 1-4 scale, and the socio-emotional index is constructed using the same method from 26 psychosocial and personality items measured on a 1-5 scale. At age 29, I measure earnings as the logarithm of one plus weekly labor income, combining cash and in-kind payments converted to weekly equivalents. All test scores and indices are standardized within the sample.

A.4 Bandwidth Selection: MSE- and CER-Optimal

I report all RD estimates using two data-driven bandwidth selectors, implemented as the `mserd` and `cerrd` options in `rdrobust`. The MSE-optimal selector (Calonico et al., 2014)

minimizes the asymptotic mean squared error of the RD point estimator and thereby delivers an optimal bias-variance trade-off for point estimation. The CER-optimal selector (Calonico et al., 2020) is designed for inference: it minimizes the coverage error of robust bias-corrected confidence intervals. The two selectors therefore target different objectives, precision of the point estimate versus accuracy of interval estimation, and by construction the CER-optimal bandwidth is narrower. I use the MSE-optimal bandwidth as the primary specification and report the CER-optimal bandwidth alongside it. The similarity of the estimates across the two selectors indicates that the results are not sensitive to the choice between them.

B Robustness to Unclustered Standard Errors

My main estimates cluster standard errors at the district level because DPEP eligibility was assigned by district and the running variable takes a single value for all individuals in a district. Clustering affects both inference and the choice of bandwidth: the data-driven selectors in `rdrobust` use the estimated variance, so treating each district as one unit leads them to select a wider window that contains enough districts on either side of the cutoff. To show how much my conclusions depend on this choice, Table A6 and Table A10 re-estimate the main specifications using heteroskedasticity-robust nearest-neighbor standard errors without clustering. All other features of the specification remain unchanged: local linear regression with a triangular kernel, sampling weights, and the mass-point adjustment, with the MSE- and CER-optimal bandwidths reselected using the unclustered variance estimator.

Without clustering, the selected bandwidths are much narrower than those in the main specification: 3.78 percentage points of 1991 female literacy for the MSE-optimal selector, compared with 11.72 in Table 4, and 1.87 for the CER-optimal selector, compared with 8.54. The estimated effects of eligibility are nonetheless close to, and slightly larger than, the main estimates. With the MSE-optimal bandwidth, eligibility raises the standardized test score by 0.158 standard deviations for all children, 0.165 for boys, and 0.148 for girls. With the CER-optimal bandwidth, the corresponding estimates are 0.195, 0.185, and 0.205 standard deviations. As in every other specification, the effects for boys and girls are similar. The standard errors, however, fall from about 0.07 to between 0.010 and 0.021, and every estimate is significant at the 1 percent level. This reduction reflects the assumption that test scores of children within a district are independent, which is not credible when eligibility and the program inputs that followed it were common to all children in the district. The unclustered standard errors therefore overstate the precision of the estimates.

The same pattern appears in the PLFS. The MSE-optimal bandwidth falls from 9.46 to 8.73, and the CER-optimal bandwidth from 6.89 to 4.97. The central result for regular wage and salaried employees is essentially unchanged in magnitude. Eligibility raises log weekly earnings among regular employees by 0.277 with the MSE-optimal bandwidth and by 0.263 with the CER-optimal bandwidth, compared with 0.281 and 0.259 in Table 5; earnings are therefore about 30 to 32 percent higher in districts just eligible for DPEP. For men, the estimates are 0.241 and 0.237, close to the clustered estimates of 0.248 and 0.223. All four estimates are significant at the 1 percent level. For women in regular employment, the MSE-optimal estimate is 0.335 and is significant at the 5 percent level, while the CER-optimal estimate, 0.210, is based on about 600 women and is not statistically significant.

The estimates for casual laborers are negative and, without clustering, significant at the 1 percent level for all persons, men, and women in both panels. I do not interpret this as stronger evidence of an effect on casual earnings. The point estimates are similar to the clustered estimates; only the standard errors have changed. Subsection 7.1 shows that the casual-labor estimates shrink toward zero and lose significance once the sampling weights are removed, and this fragility is not addressed by changing the variance estimator. The estimates for the labor force as a whole and for the self-employed remain imprecise. The exceptions are marginally significant negative estimates for self-employed men and for women in the labor force, neither of which is significant in the weighted clustered specifications.

Taken together, the unclustered estimates show that my point estimates do not depend on the choice of clustering: the effects on foundational skills and on the earnings of regular employees are similar in magnitude with and without clustering. Ignoring the district-level assignment of eligibility, however, would sharply overstate the precision of these estimates. I therefore regard the clustered standard errors in the main tables as the appropriate basis for inference.

C Foundational Skills and Early-Career Earnings: Fuzzy RD

The reduced-form results in Section 6 show that DPEP eligibility raised both foundational learning and the earnings of regular employees at the threshold, for the overall sample and for men. Subsection 8.1 combines the two in a two-sample ratio. This section estimates the same object within the PLFS sample, using eligibility as an instrument for a district-level measure of foundational skills in a fuzzy regression discontinuity design.

Each PLFS respondent is assigned the standardized ASER composite test score of his or her district, constructed from the pooled 2006-2011 rounds as described in Section 3. Let S_d denote this score in standard deviation units of the district average, $\tilde{R}_d$ the running variable, and $D_d = \mathbb{1}\{\tilde{R}_d \geq 0\}$ DPEP eligibility. I estimate

$$S_d = \pi_0 + \pi D_d + \pi_1 \tilde{R}_d + \pi_2 D_d \times \tilde{R}_d + u_d, \tag{7}$$

$$Y_{id} = \alpha + \beta_d S_d + \delta_1 \tilde{R}_d + \delta_2 D_d \times \tilde{R}_d + \varepsilon_{id}, \qquad |\tilde{R}_d| \leq h, \tag{8}$$

where Y_{id} is the log of one plus weekly earnings for individual i in district d. The estimation sample consists of regular wage and salaried employees. Equation 7 is the first stage, and π measures the discontinuity in district learning at the cutoff. Equation 8 is the second stage, estimated by two-stage least squares, with D_d as an instrument for S_d. The coefficient β_d equals the earnings discontinuity in the PLFS divided by the discontinuity in district-average learning at the cutoff, and therefore measures the change in log weekly earnings per standard deviation of the district learning score. I use local linear regression with a triangular kernel, the MSE-optimal bandwidth for the fuzzy design, PLFS sampling weights, and the mass-point adjustment in `rdrobust`. Because eligibility is assigned at the district level and the number of districts near the cutoff is modest, I report two p-values for the second stage: the district-clustered p-value from `rdrobust` and a wild cluster bootstrap

p-value with Webb weights and 999 replications (Cameron et al., 2008; Webb, 2023). I report estimates for the full sample and for men.

Table A11 reports the estimates. Panel A reports the first stage. Within the MSE-optimal bandwidth, DPEP eligibility raises the district learning score by 0.659 standard deviations of the district average in the full sample ($p = 0.045$) and by 0.664 among men ($p = 0.072$). Panel B reports the second stage. Among regular employees, a one-standard-deviation increase in the district learning score is associated with weekly earnings that are 0.424 log points higher when the score is instrumented by eligibility, an increase of about 53 percent. For men, the corresponding estimate is 0.357 log points, an increase of about 43 percent.

These coefficients are not comparable to the individual-level gradient in the Young Lives Study, because a standard deviation of the district average is much smaller than a standard deviation of an individual child's score. The sharp RD estimates imply the conversion directly: eligibility raises the individual-level composite by 0.137 standard deviations (Table 4) and the district-level score by 0.659 district standard deviations, so one district standard deviation corresponds to roughly $0.137/0.659 \approx 0.21$ individual standard deviations. Expressed per individual standard deviation, the fuzzy estimate for all persons is therefore about $0.424/0.21 \approx 2.0$ log points, close to the two-sample ratio of 2.05 in Table 6. The single-sample and two-sample approaches thus recover the same quantity, and both imply a scaled effect far larger than any plausible return to foundational skills alone.

Statistical precision depends on the inference method. The district-clustered `rdrobust` p-values for the second stage are 0.098 for all persons and 0.172 for men, so the pooled estimate is significant only at the 10 percent level and the male estimate is not statistically significant under asymptotic inference. The wild cluster bootstrap p-values are lower, at 0.027 for all persons and 0.074 for men. The first stage is significant at the 5 percent level in the full sample and only at the 10 percent level for men; with z-statistics of about 2 or below, the instrument is weak in this sample. I interpret the evidence as consistent in sign and magnitude with the reduced-form results, but I do not treat the fuzzy estimates as more precise than the sharp RD estimates in Table 5. Sampling error in the first stage adds to the uncertainty in the earnings estimate.

These estimates should be interpreted with the same caution as the reduced-form results. The instrument, DPEP eligibility, affects earnings through channels other than measured foundational learning: it raised enrollment and completed years of schooling (Azam and Saing, 2017; Khanna, 2023a) and may have changed local labor markets. The exclusion restriction required to interpret β_d as the effect of learning therefore fails, and β_d is best viewed as the earnings effect of the entire program expressed per standard deviation of district learning, an upper bound on the learning channel. Two further points qualify the estimates. First, ASER and the PLFS cover the same birth cohorts and districts but not the same individuals, so the district score measures the cohort's learning environment rather than each worker's skills. Second, the estimates condition on being a regular employee, so they combine higher productivity within these jobs with any change in who holds them.

Read this way, the fuzzy estimates serve as a consistency check rather than a new identification result. The sign and magnitude of the earnings effect do not depend on whether it is expressed per unit of eligibility or per unit of learning, or on whether the two discontinuities are estimated in one sample or two. The surrogate-index exercise in Subsection 8.2 asks the

complementary question of how much of the observed effect measured learning can account for.

Appendix Figures and Tables

Table A1: Foundational Skills and Earnings at Age 29, by Gender (YLS)

	(1)	(2)	(3)
	Log Weekly Income (29 yrs)		
	Pooled	Male	Female
Reading Score (8 yrs)	0.368***	0.387***	0.217
	(0.122)	(0.135)	(0.310)
Controls	Yes	Yes	Yes
YLS Cluster FE	Yes	Yes	Yes
Observations	633	381	252
R^2	0.196	0.163	0.142

Notes: Standard errors clustered at the sentinel-site (YLS cluster) level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All specifications include cluster fixed effects. Column (1) reports the pooled estimate, which also controls for gender. Columns (2) and (3) estimate the same specification separately for males and females. Controls include the child's age in months, household size, caste (Scheduled Tribe, Backward Class, Other Castes; omitted category Scheduled Caste), religion (Muslim, Buddhist, Hindu, Sikh; omitted category Christian), an indicator for the caregiver not having completed primary schooling, and the household wealth index. Coefficients on the controls are not reported. In the male sample, the Sikh indicator is omitted because no observations fall in that category. The sample comprises children who were 8 years old in the first wave of the Young Lives Survey in India in 2002 and were followed up to age 29 (2023-24). The Reading Score measures foundational literacy at age 8 (Round 1) and is standardized. Earnings are measured as $\ln(1 + \text{weekly labor income})$, where income combines cash and in-kind payments converted to weekly equivalents, at age 29 (Round 7).

Table A2: Distribution of Foundational Skill Levels, Children Aged 8-11

		Proportion (%)					
Level	Description	2006	2007	2008	2009	2010	2011
Panel A: Reading							
0	Cannot read	8.59	5.13	5.72	4.77	5.01	6.71
1	Letters	12.06	12.60	13.44	13.55	12.99	16.07
2	Words	19.32	20.24	18.49	20.09	20.34	20.06
3	Paragraph	24.65	23.00	23.91	24.74	24.37	21.85
4	Story	35.39	39.02	38.44	36.85	37.29	35.31
	Observations	227,165	237,003	224,463	213,163	200,064	186,282
Panel B: Arithmetic							
0	Cannot recognise numbers	13.80	5.39	5.42	4.88	4.92	5.99
1	One-digit number recognition	-	13.32	15.54	14.32	14.55	18.79
2	Two-digit number recognition	26.29	24.66	26.13	25.55	26.70	27.89
3	Subtraction	30.28	29.60	28.01	29.30	29.49	28.02
4	Division	29.63	27.02	24.91	25.95	24.33	19.30
	Observations	225,452	236,299	224,052	211,799	198,655	185,132

Notes: The table reports the percentage distribution of children aged 8-11 across foundational reading and arithmetic skill levels in ASER from 2006 to 2011. Reading levels range from 0 (cannot read) to 4 (can read a story), while arithmetic levels range from 0 (cannot recognize numbers) to 4 (can solve division). One-digit number recognition was not assessed in 2006; therefore, the corresponding entry is not applicable (shown as -).

Table A3: Robustness: RD Estimates of Effect on Standardized Test Scores (Unweighted)

	All	Boys	Girls
	(1)	(2)	(3)
Panel A: MSE-optimal bandwidth			
RD estimate	0.185**	0.184**	0.186**
	(0.081)	(0.084)	(0.079)
Bandwidth (h)	8.512	8.512	8.512
Bias bandwidth (b)	13.552	13.552	13.552
Effective observations	364,793	195,894	168,899
Observations	1,298,036	703,826	594,210
Clusters (districts)	299	299	299
Panel B: CER-optimal bandwidth			
RD estimate	0.164*	0.157*	0.171*
	(0.089)	(0.092)	(0.087)
Bandwidth (h)	6.204	6.204	6.204
Bias bandwidth (b)	13.552	13.552	13.552
Effective observations	284,194	152,555	131,639
Observations	1,298,036	703,826	594,210
Clusters (districts)	299	299	299

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The dependent variable is the standardized test score, constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively. Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust without sampling weights and adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Robustness: RD Estimates on Test Scores (Local Quadratic)

	All	Boys	Girls
	(1)	(2)	(3)
Panel A: With sampling weights			
RD estimate	0.160*	0.167*	0.151*
	(0.087)	(0.088)	(0.089)
Bandwidth (h)	13.916	13.916	13.916
Bias bandwidth (b)	19.352	19.352	19.352
Effective observations	665,670	359,572	306,098
Observations	1,298,036	703,826	594,210
Clusters (districts)	415	415	415
Panel B: Without sampling weights			
RD estimate	0.195**	0.192**	0.197**
	(0.092)	(0.094)	(0.089)
Bandwidth (h)	13.700	13.700	13.700
Bias bandwidth (b)	19.039	19.039	19.039
Effective observations	655,451	353,915	301,536
Observations	1,298,036	703,826	594,210
Clusters (districts)	407	407	407

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The dependent variable is the standardized test score, constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively. Sharp RD estimates from local quadratic regression with a triangular kernel, estimated with rdrobust with ASER sampling weights (Panel A) and without sampling weights (Panel B), adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Robustness: Fuzzy RD Estimates on Test Scores (Add Four DPEP Districts)

	All	Boys	Girls
	(1)	(2)	(3)
Panel A: First stage, with sampling weights			
Discontinuity in DPEP	0.743***	0.745***	0.742***
	(0.107)	(0.106)	(0.108)
Panel B: Fuzzy RD estimate, with sampling weights			
DPEP	0.185*	0.189*	0.180*
	(0.099)	(0.099)	(0.103)
Bandwidth (h)	11.722	11.722	11.722
Bias bandwidth (b)	18.554	18.554	18.554
Effective observations	550,344	296,493	253,851
Observations	1,298,036	703,826	594,210
Clusters (districts)	399	399	399
Panel C: First stage, without sampling weights			
Discontinuity in DPEP	0.730***	0.734***	0.725***
	(0.100)	(0.098)	(0.101)
Panel D: Fuzzy RD estimate, without sampling weights			
DPEP	0.253**	0.250**	0.256**
	(0.122)	(0.124)	(0.121)
Bandwidth (h)	8.512	8.512	8.512
Bias bandwidth (b)	13.552	13.552	13.552
Effective observations	364,793	195,894	168,899
Observations	1,298,036	703,826	594,210
Clusters (districts)	299	299	299

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The dependent variable is the standardized test score, constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively. Fuzzy RD estimates from local linear regression with a triangular kernel, estimated with rdrobust adjusting for mass points in the running variable, with ASER sampling weights (Panels A and B) and without sampling weights (Panels C and D). The treatment indicator DPEP equals one for districts below the national average and for four districts above the national average that were also covered under DPEP. The first stage reports the discontinuity in DPEP at the cutoff. Bandwidths are the MSE-optimal bandwidths from the corresponding sharp RD specification. The running variable is the negative deviation of the 1991 district female literacy from the national average. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Robustness: RD Estimates of on Standardized Test Scores (Without Clustering)

	All (1)	Boys (2)	Girls (3)
Panel A: MSE-optimal bandwidth			
RD estimate	0.158***	0.165***	0.148***
	(0.010)	(0.014)	(0.015)
Bandwidth (h)	3.784	3.784	3.784
Bias bandwidth (b)	8.016	8.016	8.016
Effective observations	186,792	99,929	86,863
Observations	1,298,036	703,826	594,210
Panel B: CER-optimal bandwidth			
RD estimate	0.195***	0.185***	0.205***
	(0.014)	(0.020)	(0.021)
Bandwidth (h)	1.872	1.872	1.872
Bias bandwidth (b)	8.016	8.016	8.016
Effective observations	104,971	56,156	48,815
Observations	1,298,036	703,826	594,210

Notes: The sample consists of children aged 8-11 surveyed in ASER rounds 2006 to 2011. The dependent variable is the standardized test score, constructed from the sum of two binary indicators for the ability to do division and the ability to read a story, the highest skills in maths and reading, respectively. Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust using ASER sampling weights and adjusting for mass points in the running variable. Bandwidths are reselected without clustering. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Heteroskedasticity-robust nearest-neighbor standard errors, not clustered, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Robustness: RD Estimates on Earnings (Unweighted)

	All persons				Males				Females			
	All workers (1)	Self-employed (2)	Regular (3)	Casual (4)	All workers (5)	Self-employed (6)	Regular (7)	Casual (8)	All workers (9)	Self-employed (10)	Regular (11)	Casual (12)
Panel A: MSE-optimal bandwidth												
RD estimate	-0.033	-0.010	0.199**	-0.025	0.028	-0.061	0.168**	-0.041	-0.445	-0.091	0.249*	0.015
	(0.232)	(0.472)	(0.089)	(0.071)	(0.227)	(0.470)	(0.083)	(0.065)	(0.364)	(0.619)	(0.143)	(0.133)
Bandwidth (h)	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002
Bias bandwidth (b)	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407
Effective observations	26,866	11,192	5,665	5,634	20,309	7,727	4,606	4,874	6,555	3,463	1,059	760
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	324	324	324	318	324	324	323	318	324	319	267	217
Panel B: CER-optimal bandwidth												
RD estimate	-0.129	-0.329	0.171*	-0.079	-0.055	-0.384	0.132	-0.093	-0.683*	-0.495	0.214	-0.044
	(0.255)	(0.542)	(0.101)	(0.080)	(0.253)	(0.535)	(0.094)	(0.071)	(0.407)	(0.724)	(0.158)	(0.158)
Bandwidth (h)	7.284	7.284	7.284	7.284	7.284	7.284	7.284	7.284	7.284	7.284	7.284	7.284
Bias bandwidth (b)	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407
Effective observations	20,196	8,216	4,431	4,147	15,322	5,690	3,612	3,618	4,872	2,524	819	529
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	324	324	324	318	324	324	323	318	324	319	267	217

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. The dependent variable is the log of weekly real earnings, where weekly earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices. Columns are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust without sampling weights and adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Robustness: RD Estimates on Earnings (Local Quadratic)

	All persons				Males				Females			
	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: With sampling weights												
RD estimate	-0.195	-0.312	0.299**	-0.181*	-0.108	-0.604	0.257**	-0.105	-0.660	0.001	0.356	-0.413*
	(0.333)	(0.662)	(0.138)	(0.105)	(0.369)	(0.668)	(0.125)	(0.083)	(0.584)	(0.910)	(0.233)	(0.214)
Bandwidth (h)	12.532	12.532	12.532	12.532	12.532	12.532	12.532	12.532	12.532	12.532	12.532	12.532
Bias bandwidth (b)	16.582	16.582	16.582	16.582	16.582	16.582	16.582	16.582	16.582	16.582	16.582	16.582
Effective observations	38,565	16,264	7,872	8,000	29,147	11,080	6,502	6,952	9,416	5,182	1,370	1,048
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	357	357	357	350	357	357	356	350	357	352	293	240
Panel B: Without sampling weights												
RD estimate	-0.195	-0.641	0.144	-0.132	-0.084	-0.717	0.105	-0.148	-0.972*	-0.907	0.106	-0.091
	(0.309)	(0.699)	(0.124)	(0.107)	(0.308)	(0.685)	(0.118)	(0.093)	(0.503)	(0.942)	(0.181)	(0.197)
Bandwidth (h)	9.820	9.820	9.820	9.820	9.820	9.820	9.820	9.820	9.820	9.820	9.820	9.820
Bias bandwidth (b)	14.279	14.279	14.279	14.279	14.279	14.279	14.279	14.279	14.279	14.279	14.279	14.279
Effective observations	26,304	10,874	5,607	5,519	19,888	7,509	4,559	4,766	6,414	3,363	1,048	753
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	313	313	313	307	313	313	312	307	313	308	260	211

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. The dependent variable is the log of weekly real earnings, where weekly earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices. Columns are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). Sharp RD estimates from local quadratic regression with a triangular kernel, estimated with rdrobust with PLFS sampling weights (Panel A) and without sampling weights (Panel B), adjusting for mass points in the running variable. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Robustness: Fuzzy RD Estimates on Earnings (Add Four DPEP Districts)

	All persons				Males				Females			
	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual	All workers	Self-employed	Regular	Casual
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: First stage, with sampling weights												
Discontinuity in DPEP	0.744***	0.739***	0.810***	0.697***	0.740***	0.722***	0.811***	0.714***	0.754***	0.772***	0.808***	0.592***
	(0.114)	(0.130)	(0.093)	(0.133)	(0.113)	(0.125)	(0.091)	(0.129)	(0.125)	(0.152)	(0.116)	(0.182)
Panel B: Fuzzy RD estimate, with sampling weights												
DPEP	-0.231	-0.306	0.347***	-0.229*	-0.157	-0.709	0.306**	-0.147	-0.683	0.180	0.417*	-0.565
	(0.357)	(0.679)	(0.130)	(0.134)	(0.402)	(0.708)	(0.121)	(0.108)	(0.575)	(0.950)	(0.228)	(0.346)
Bandwidth (h)	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459	9.459
Bias bandwidth (b)	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259	14.259
Effective observations	24,805	10,204	5,289	5,221	18,858	7,108	4,326	4,499	5,945	3,094	963	722
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	313	313	313	307	313	313	312	307	313	308	260	211
Panel C: First stage, without sampling weights												
Discontinuity in DPEP	0.798***	0.826***	0.801***	0.758***	0.790***	0.810***	0.811***	0.762***	0.819***	0.854***	0.767***	0.729***
	(0.091)	(0.090)	(0.092)	(0.105)	(0.093)	(0.093)	(0.087)	(0.102)	(0.087)	(0.090)	(0.126)	(0.143)
Panel D: Fuzzy RD estimate, without sampling weights												
DPEP	-0.042	-0.012	0.249**	-0.033	0.036	-0.076	0.207**	-0.054	-0.542	-0.106	0.325*	0.020
	(0.289)	(0.571)	(0.104)	(0.094)	(0.288)	(0.576)	(0.095)	(0.088)	(0.434)	(0.720)	(0.181)	(0.182)
Bandwidth (h)	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002	10.002
Bias bandwidth (b)	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407	15.407
Effective observations	26,866	11,192	5,665	5,634	20,309	7,727	4,606	4,874	6,555	3,463	1,059	760
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Clusters (districts)	324	324	324	318	324	324	323	318	324	319	267	217

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. The dependent variable is the log of weekly real earnings, where weekly earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices. Columns are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). Fuzzy RD estimates from local linear regression with a triangular kernel, estimated with rdrobust adjusting for mass points in the running variable, with PLFS sampling weights (Panels A and B) and without sampling weights (Panels C and D). The treatment indicator DPEP equals one for districts below the national average and for four districts above the national average that were also covered under DPEP. The first stage reports the discontinuity in DPEP at the cutoff. Bandwidths are the MSE-optimal bandwidths from the corresponding sharp RD specification. The running variable is the negative deviation of the 1991 district female literacy from the national average. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Robustness: RD Estimates of Effect Earnings (Without Clustering)

	All persons				Males				Females			
	All workers (1)	Self-employed (2)	Regular (3)	Casual (4)	All workers (5)	Self-employed (6)	Regular (7)	Casual (8)	All workers (9)	Self-employed (10)	Regular (11)	Casual (12)
Panel A: MSE-optimal bandwidth												
RD estimate	-0.184	-0.275	0.277***	-0.173***	-0.124	-0.559*	0.241***	-0.114***	-0.557*	0.059	0.335**	-0.355***
	(0.151)	(0.238)	(0.060)	(0.038)	(0.167)	(0.291)	(0.072)	(0.035)	(0.307)	(0.371)	(0.163)	(0.098)
Bandwidth (h)	8.729	8.729	8.729	8.729	8.729	8.729	8.729	8.729	8.729	8.729	8.729	8.729
Bias bandwidth (b)	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069
Effective observations	23,513	9,690	5,013	4,972	17,838	6,739	4,098	4,277	5,673	2,949	915	695
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720
Panel B: CER-optimal bandwidth												
RD estimate	-0.190	-0.328	0.263***	-0.244***	-0.082	-0.632*	0.237***	-0.145***	-0.710*	-0.129	0.210	-0.464***
	(0.202)	(0.315)	(0.064)	(0.053)	(0.224)	(0.384)	(0.076)	(0.046)	(0.409)	(0.497)	(0.254)	(0.119)
Bandwidth (h)	4.974	4.974	4.974	4.974	4.974	4.974	4.974	4.974	4.974	4.974	4.974	4.974
Bias bandwidth (b)	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069	13.069
Effective observations	14,070	5,592	3,223	2,839	10,569	3,801	2,620	2,488	3,499	1,789	603	351
Observations	76,788	34,556	14,788	15,171	57,759	23,287	12,141	13,451	19,025	11,265	2,647	1,720

Notes: The sample consists of individuals aged 22-25 surveyed in PLFS rounds 2020 to 2025. The dependent variable is the log of weekly real earnings, where weekly earnings are the sum of daily earnings from wage activities over the reference week plus monthly regular salary and self-employment earnings scaled to a week, deflated to constant prices. Columns are defined by current weekly status: all workers (codes 11-81), self-employed (codes 11-21), regular salaried (code 31), and casual laborers (codes 41-51). Sharp RD estimates from local linear regression with a triangular kernel, estimated with rdrobust using PLFS sampling weights and adjusting for mass points in the running variable. Bandwidths are reselected without clustering. The running variable is the negative deviation of 1991 district female literacy from the national average, so the estimate is the discontinuity for districts below the cutoff. Heteroskedasticity-robust nearest-neighbor standard errors, not clustered, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Foundational skills and Earnings of Regular Employees

	All persons (1)	Males (2)
Panel A: First stage. Dependent variable: Z test score (district)		
DPEP eligibility	0.659**	0.664*
p-value (rdrobust)	[0.045]	[0.072]
Panel B: Second stage. Dependent variable: log weekly earnings		
Z Test Score (district)	0.424*	0.357
p-value (rdrobust)	[0.098]	[0.172]
p-value (wild cluster bootstrap)	[0.027]	[0.074]
Bandwidth (h)	9.737	8.487
Observations within bandwidth	5,576	4,024

Notes: The sample consists of regular wage and salaried employees aged 22-25 in rural India, drawn from PLFS rounds 2020 to 2025. In Panel A, the dependent variable is the standardized ASER composite test score for the individual's district. In Panel B, the dependent variable is the log of one plus weekly earnings, and the district test score is instrumented by DPEP eligibility, defined as an indicator for a 1991 district female literacy rate below the national average. The running variable is the negative deviation of 1991 district female literacy from the national average. Estimates are from a fuzzy RD with local linear regression, a triangular kernel, an MSE-optimal bandwidth, PLFS sampling weights, and a mass-point adjustment. The 2SLS point estimate is computed using the product of kernel and sampling weights, with separate slopes on each side of the cutoff. The wild cluster bootstrap uses Webb weights and 999 replications. p-values are in brackets. Standard errors clustered at the district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.